

Modern Force and Coercive Backfire

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ABSTRACT. Modern military technologies, from automated weapons to AI-assisted precision strikes, promise cheaper force. Do these alternatives reduce the costs of interstate conflict? The key is how opponents respond. I develop a parsimonious bargaining framework of conflict where a state gains a *subwar capability*: a cheaper operation short of general war. This option can replace war after resistance or settlement after compliance, potentially weakening both threats and assurances in coercive bargaining. The target may concede less, reducing the state's incentive for restraint and making war more attractive. Comparing the coercer's ability to secure concessions with and without the option, I show this response can lower coercive ability and increase war even when the operation is never used. Subwar capability can weaken the effect of military power on coercive ability and reduce social welfare by raising conflict costs. More efficient military technology can disrupt coercive bargaining and make interstate conflict more costly.

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INTRODUCTION

Violence has declined over the long run, argues [Pinker \(2011\)](#). In interstate politics, [Altman \(2020\)](#) documents a shift from wars for large territories toward small seizures designed to avoid war. Automated weapons and AI-assisted precision strikes promise to make military operations less costly still ([Zegart 2018](#); [Jensen, Whyte, and Cuomo 2019](#); [Horowitz 2020](#)). In March 2024, U.S. Deputy Secretary of Defense Kathleen Hicks argued that better decisions enabled by AI could be “decisive in deterring a fight, and winning a fight.”^[1] In April 2026, Danylo Tsvok, head of Ukraine’s Defense AI Center A1, similarly linked defensive capability to peace: “Democracies must develop strong defensive capabilities,” he said. “Without AI, they cannot effectively protect peace.”^[2] I call access to a cheaper operation short of general war a *subwar capability*. The appeal is that such operations could reduce the costs of interstate conflict by substituting for general war.

I show that subwar capability can weaken a government’s ability to secure concessions without war and raise the overall cost of interstate conflict. The subwar option can change what the target expects from resisting or complying, and therefore how much it is willing to concede. Coercion requires both a credible *threat* to punish resistance and a credible *assurance* that compliance will prevent further harm ([Schelling 1960](#)). The subwar option can weaken both: it makes resistance less dangerous when it replaces a more damaging war, and compliance less protective when it replaces settlement. Either change reduces what the target will concede. Smaller concessions reduce the demanding government’s incentive for restraint and can make general war more attractive, even when the subwar operation is never used. Subwar operations can also add costs when they replace settlement.

I develop this argument in a parsimonious bargaining model of conflict in which one state, the coercer, demands a concession, the target complies or resists, and the coercer then chooses whether to use force ([Banks 1990](#); [Fearon 1994](#); [Schultz 1999](#)). The target does not know how costly the coercer would find a war. I compare a benchmark with settlement

^[1]Kathleen Hicks, “[AI in the Era of Strategic Competition](#),” March 5, 2024.

^[2]Derek Gatopoulos and Hanna Arhirova, [interview with Danylo Tsvok](#), Associated Press, April 2026.

and general war to a game that adds the subwar option after either response. Following the experimental approach to formal modeling in [Paine and Tyson \(2020\)](#), I vary only the availability of the subwar option, holding all other features of the bargaining environment fixed. In each game, I select the equilibrium with the largest accepted demand. I measure *coercive ability* as the expected concession retained without general war: what the target gives up, discounted by the risk that war reopens the dispute. I also assess the coercer's total payoff, the probability of war, and the two states' combined conflict costs.

This bargaining mechanism can weaken the translation of military power into coercive ability. Without the subwar option, greater military power makes resistance more dangerous, so the target is willing to concede more to avoid war. That larger concession also gives the coercer more to preserve through settlement, reducing its incentive to fight. A subwar operation can disrupt this exchange. The coercer keeps the target's concession even if it conducts the operation afterward. An extra concession therefore raises the payoff from stopping and from operating by the same amount; it does not make stopping more attractive relative to the operation. The target consequently cannot count on a larger concession to remove this source of further harm. Its willingness to concede depends on how much safer compliance makes it, as well as how dangerous resistance becomes. When the subwar option erodes that protection, the target may respond less to an increase in military power. The resulting smaller concession gives the coercer less to preserve through restraint and can make war more attractive after compliance. Greater military power can therefore produce a smaller increase in the concession the coercer expects to retain without war.

The resulting reduction in concessions can also increase the risk of general war, despite the availability of a cheaper substitute. With the target making the same concession or resisting as before, the coercer may choose a subwar operation where it would otherwise wage general war. Anticipating this less damaging response to resistance, the target may be willing to concede less. The prospect of a subwar operation even after compliance can further reduce its willingness to concede. A smaller concession leaves the coercer less to preserve through restraint, making general war more attractive. If the concession falls

sufficiently, this bargaining response outweighs the direct substitution away from war. The subwar option can have the immediate effect of replacing war with limited force, yet the ultimate effect of making war more likely by undermining coercive threats and assurances, thereby inducing smaller concessions.

These changes in bargaining and the use of force can also raise the two states' combined conflict costs, even as subwar operations become cheaper. Replacing war with a subwar operation saves costs, while replacing settlement adds them. When the added costs exceed the savings, cheaper operations increase expected conflict costs even before the concession adjusts. When cheaper operations also reduce the target's concession, that bargaining response makes war more attractive and adds further costs. Evaluating a capability only by the expense of each operation or the number of general wars can therefore miss the costs it creates by changing bargaining and the use of force.

These findings contribute to research on the benefits and risks of force short of war. Limited seizures can secure gains without provoking war (Tarar 2016; Altman 2017; 2020; Maass 2021), while low-level operations can avert preventive war by slowing an adversary's rise (Schram 2021). Other work identifies risks: less destructive retaliation can make nuclear exchanges more likely (Powell 1989), and improved capabilities to counter low-level aggression can induce riskier challenges and war (Gannon et al. 2024). Schram (2022) shows that better low-level capabilities can leave every type of their holder no better off through greater predictability or increased risk taking by its opponent. Spaniel and İdrisoglu (2024) show that switching to a cheaper military strategy can reduce demands and increase war risk; with shifting power, both states can lose. In their baseline, acceptance ends the dispute and war follows rejection. Here, the subwar option remains available after both resistance and compliance. Weaker punishment or protection can reduce the concession, leaving the coercer less to preserve through restraint. Under the baseline conditions, I identify the concession-loss threshold at which no coercer type gains and beyond which war after compliance becomes more likely. This connects capability backfire to war after compliance even when no type uses the added operation in equilibrium.

The paper also contributes to research on the relationship between military power and coercive ability. The capacity to impose costs need not track the capacity to win a war (Slantchev 2003; 2011; Kenkel and Schram 2025). Sechser (2010) shows that weaker targets may resist militarily superior coercers because compliance can reveal low resolve and invite further demands. More generally, powerful coercers can fail when they cannot credibly promise restraint (Art and Greenhill 2018; Cebul, Dafoe, and Monteiro 2021). Here, the coercer retains the concession even if it conducts a subwar operation, so a larger concession does not make settlement more attractive relative to operating. I show that this assurance problem can make an increase in military power produce a smaller gain in expected concessions retained without war when the subwar option is available.

Finally, the model explains how the terms of a bargain connect threats and assurances. Coercion requires both costly resistance and protection through compliance (Schelling 1966; Cebul, Dafoe, and Monteiro 2021; Pauly 2024; 2025). Pape (1996) classifies strategies by how they make resistance costly or futile. Kydd and McManus (2017) show that threats and assurances can both be credible and that anticipated attack after a concession can reduce what the target will concede. Ioku and Kurizaki (2026) show how the prospect of attack after compliance can allow coercers planning war to imitate the demands of those seeking settlement. Adding the same subwar option after either response can connect threats and assurances through the concession. Even when the operation only weakens punishment for resistance, the target can concede less, making war more attractive after compliance. I thus identify a channel through which weaker threats undermine assurance by reducing what the coercer has to preserve through restraint.

THE FRAMEWORK

Figure 1 shows the common timing. The coercer makes a take-it-or-leave-it demand, the target concedes or resists, and the coercer chooses a continuation action. The coercer's cost of war is private information, and abandoning a resisted demand carries an audience cost

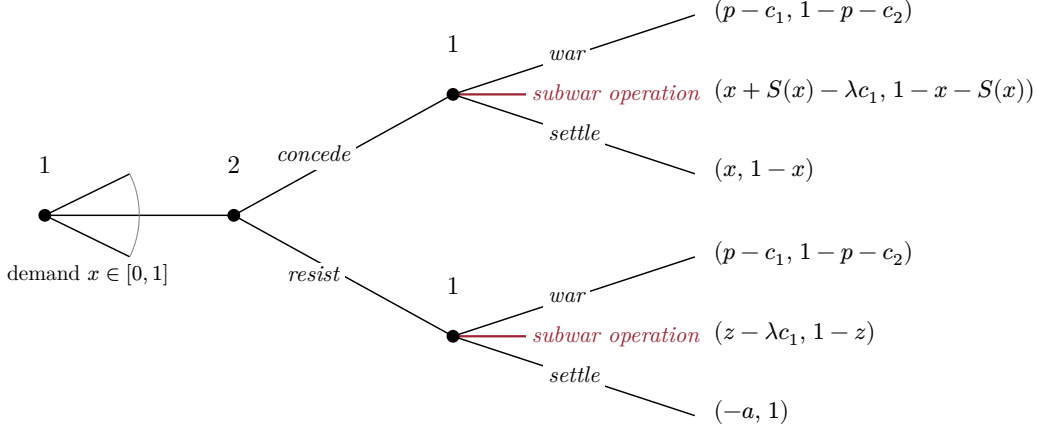


Figure 1: Coercion Game

Note: The tree is conditional on the coercer's privately observed cost c_1 . Player 1 is the coercer and player 2 the target, which observes the demand but knows only the cost distribution; the coercer's payoff appears first. The colored subwar-operation branches are added by the subwar capability and absent from the benchmark. War reopens the dispute, so its payoffs do not depend on the transfer. A subwar operation preserves the transfer, produces gain $S(x) = \min\{z, 1 - x\}$ after compliance and z after resistance, and costs λc_1 . Settling after resistance costs the coercer a .

(Banks 1990; Fearon 1994; Schultz 1999). The benchmark permits settlement or war; the second game adds a subwar operation after either target response.

Players, Actions, and Timing

Two states value a disputed good at 1. The *target*, player 2, initially holds the good. The *coercer*, player 1, has a war cost c_1 drawn uniformly from $[0, \bar{c}]$. The coercer observes its cost, but the target knows only its distribution F , with density $f = \frac{1}{\bar{c}}$ on its support. Throughout, $F(t) = 0$ for $t \leq 0$ and $F(t) = 1$ for $t \geq \bar{c}$. A lower cost denotes a more resolute coercer. The target's war cost $c_2 \geq 0$ is common knowledge.

The coercer demands $x \in [0, 1]$. The target chooses $r \in \{0, 1\}$, where $r = 1$ denotes resistance and $r = 0$ compliance. Let y be the transfer in place at the continuation stage. Thus $y = x$ after compliance and $y = 0$ after resistance. The coercer chooses from the same

continuation menu after either response. A continuation history is (y, r) ; compliance with a zero demand is $(0, 0)$.

Settlement leaves the transfer y in place. If the target resisted, settlement also imposes an audience cost $a > 0$ on the coercer for abandoning its demand. War gives payoffs $(p - c_1, 1 - p - c_2)$, where $p \in (0, 1)$ is the coercer's probability of victory. Because war reopens the whole dispute, these payoffs do not depend on any earlier transfer. At continuation indifference the coercer chooses war before the operation and the operation before settlement.

The second game adds a *subwar operation* with reach $z > 0$. The baseline puts different operations on a common payoff scale. Let

$$S(y) = \min\{z, 1 - y\} \tag{1}$$

denote the resulting gain to the coercer and harm to the target. The operation costs the coercer λc_1 , where $\lambda \in (0, 1)$, and produces payoffs $(y + S(y) - \lambda c_1, 1 - y - S(y))$.

The analysis maintains the following parameter region.

ASSUMPTION 1 (Parameter Restrictions): (i) The operation has limited reach,

$$p + c_2 + 2z \leq 1$$

and (ii) uncertainty about the coercer's war cost is sufficiently large,

$$\frac{2p + c_2}{1 - \lambda} < \bar{c}$$

A war costs the target $p + c_2$ relative to keeping the disputed good. [Assumption 1\(i\)](#) ensures that the operation has its full effect at every accepted demand, so $S(y) = z$ there.^[3] [Assumption 1\(ii\)](#) implies $\bar{c} > 2p + c_2$ in the benchmark and directly imposes the stronger

^[3]If $x > 1 - z$, the operation pays $1 - \lambda c_1 > p - c_1$ for every type, so no war follows compliance. The target then loses at least x , whereas its loss after resistance is at most $\max\{p + c_2, z\} \leq 1 - z$ under every posterior. Such a demand is therefore strictly rejected. This argument does not depend on pooling or equilibrium selection.

bound needed in the operation game. These inequalities make the target’s payoff from concession decrease with the demand under the prior. Additional conditions governing interior cutoffs and the coercer’s continuation menu are stated with the results that use them.

Comments on the Model

The comparison isolates how a subwar capability changes punishment after resistance and protection after compliance. Following [Paine and Tyson \(2020\)](#), I interpret the two games as an ideal experiment: adding the operation changes the coercer’s military options while holding the political environment and information fixed. In each game, I select the equilibrium in which the coercer makes the largest demand the target accepts. [§ Scope of the Equilibrium Comparison](#) explains which conclusions depend on that selection.

Force remains available after compliance. A concession changes the coercer’s incentives but does not commit it to restraint. Allowing a military choice after either response makes assurance part of the same bargaining problem as the threat ([Kydd and McManus 2017](#)). Both games share this commitment problem; the operation changes the incentives to honor compliance. The timing treats the target’s response as completed before the military choice and abstracts from subsequent target retaliation, renewed bargaining, and changes in decision makers.

The operation preserves the concession and pursues a residual objective. The payoff specification applies when compliance settles one part of the dispute while leaving another military objective valuable. For example, suppose the target withdraws from a contested area but leaves a separate military installation operational. A strike can then disable that installation without reversing the withdrawal. The fixed incremental gain represents an objective whose value is unchanged across the concessions considered. If compliance instead removes the installation or otherwise exhausts that objective, the fixed-gain specification

no longer applies.^[4]

Operating cost, operational gain, and war-fighting power are distinct. A reduction in λ makes a given operation cheaper for the coercer while holding its effect z and the probability of victory p fixed. Proportional costs preserve the same ranking of coercer types across the two military actions. The baseline treats the coercer’s operational gain as an equal loss to the target. [Corollary 1](#) allows the two to differ: on the active operation branch, the coercer’s gain governs its willingness to settle, while the target’s harm affects its willingness to concede.

Backing down after resistance carries a political cost. The audience cost a makes settlement less attractive after resistance; compliance removes that penalty without guaranteeing restraint. This difference supports positive concessions in the selected comparison. If $a = 0$, neither game supports a positive selected demand under the remaining parameter restrictions.^[5] The baseline assigns no audience cost to the operation; [SI §F.3. The Operation’s Audience Cost](#) derives the conditions when it carries a political cost of its own.

War reopens the entire dispute. The baseline holds the war payoff fixed across concessions, so a larger transfer improves settlement relative to war. [SI §F.2. Power Shift and Partial Recontestation](#) allows a concession to improve subsequent war-fighting prospects or remain partly secure during war. Payoff loss and increased war remain possible under the extension’s conditions, but their thresholds differ. The common threshold in [Proposition 5](#) is therefore specific to the baseline payoff structure.

The operation does not trigger general war in the baseline. This separates the bargaining consequences of subwar capability from the subwar operation’s escalation risk. [SI §F.1.](#)

^[4]The withdrawal example is hypothetical. Operation Midnight Hammer provides operational motivation for a bounded strike against fortified nuclear facilities. For its technology, see [Air Force Global Strike Command \(2026\)](#) and [U.S. Strategic Command \(2025\)](#). These accounts concern the military operation rather than a preceding accepted concession.

^[5]Resistance and compliance with zero then have identical continuation incentives. The prior acceptance margin is zero at a zero demand and strictly decreases thereafter. This boundary is distinct from the stronger restriction used in the closed-form operation analysis to rule out settlement after resistance.

[Escalation Risk](#) derives conditions under which strict reductions in coercive ability and increases in modeled conflict cost survive sufficiently small risk. Every-type payoff loss and increased war also survive under a stronger concession-loss condition.^[6] Positive escalation risk nevertheless makes settlement at the operation margin respond to the concession, so the exact constant-cutoff result changes.

Capability acquisition and prior signals are held fixed. The comparison conditions on possession and the information available before the demand. It identifies how the target responds to the subwar option. Proposition 6 evaluates the coercer’s payoff conditional on possession; an acquisition decision would also require acquisition costs and the bargaining opportunities in which the capability could be used. Pre-demand military signaling can instead change beliefs about the coercer ([Post 2019](#)). Holding that information fixed separates this channel from the change in continuation incentives studied here.

Solution Concept

I solve for pure-strategy perfect Bayesian equilibria satisfying the intuitive criterion ([Cho and Kreps 1987](#)). The target responds to each demand by either complying or resisting, without randomizing, and concedes when indifferent. After an unused demand, the criterion excludes a coercer type if every sequentially rational target response gives it strictly less than its equilibrium payoff, provided some type remains. Types that can match their equilibrium payoff remain admissible. [SI §C.1. Off-Path Beliefs and the Intuitive Criterion](#) constructs supporting beliefs. Unless a result explicitly considers the broader class of perfect Bayesian equilibria, all equilibrium claims impose the criterion.

Pooling represents every equilibrium payoff and military outcome attainable with pure target responses. Any coercer type that strictly benefits from compliance chooses the largest demand the target accepts. Only types indifferent between compliance and resistance can make other demands, and their different signals leave military actions and payoffs

^[6]The baseline must lie strictly within the active, interior region specified in the extension. The joint payoff-loss and war comparison additionally requires $x^\dagger > x^* + z$.

unchanged. SI [Lemma 5](#) proves that each such equilibrium has a pooling counterpart with the same military action and both states' payoffs for every coercer type.^[7]

I select the largest demand accepted under the prior in each game because it gives every coercer type its highest equilibrium payoff and maximizes the unconditional retained concession. It also minimizes unconditional war and expected joint conflict cost within the pure-response class, as SI [Corollary 2](#) shows: a larger concession makes settlement and the operation more attractive relative to war. Thus, an increase in war in the selected comparison raises the minimum war probability attainable in this class. The intuitive criterion does not uniquely select this equilibrium; [§ Scope of the Equilibrium Comparison](#) explains the remaining selection and refinement limits.

THE BENCHMARK

Begin with the benchmark, which excludes the subwar operation. The coercer can choose only settlement or war after either target response. I compare the games in terms of the coercer's *coercive ability*, defined as follows.

DEFINITION 1 (Coercive Ability): A coercer's *coercive ability* is the expected concession retained without general war:

$$\Omega = x [1 - \Pr(\text{war})]$$

where x is the conceded demand and $\Pr(\text{war})$ is the probability of war after compliance.

This measure excludes additional gains from the operation and continuation costs. Proposition 6 reports the coercer's expected total payoff. I compare the largest accepted demands under the common selection stated above. [§ Scope of the Equilibrium Comparison](#) discusses separating demands, alternative equilibria, and refinement limits. A dagger marks bench-

^[7]This equivalence concerns payoffs and unconditional outcomes. Different signals can change outcomes conditional on compliance; the reported coercive-ability measure uses the pooling assessment.

mark quantities. Unmarked symbols refer to the game with the operation. The Supporting Information contains all proofs.

At a fixed demand, the operation cannot mechanically reduce coercive ability. If it replaces war after compliance, the transfer survives more often. If it replaces settlement, the transfer remains in place. Any reduction in coercive ability must therefore arise because the subwar capability changes the demand the target accepts.

To solve the benchmark, start with the coercer's choice after the target responds. War becomes less attractive as its cost rises. After compliance, the coercer fights when its war cost is at or below $\kappa^{\dagger(x)} = p - x$ and settles otherwise. A larger concession lowers this cutoff because settlement preserves the transfer whereas war does not. After resistance, backing down instead carries the audience cost, raising the war cutoff to $p + a$. The full rule for either response is given in SI [Lemma 1](#).

Concession weakly raises every coercer type's continuation value. A type that settles receives x after concession instead of paying a after resistance. A type that fights receives $p - c_1$ after either response and is indifferent. Thus no type prefers resistance, and every settling type prefers the largest accepted demand.

I therefore consider a pooling equilibrium in which the demand conveys no information about c_1 . Indifferent fighting types can make other demands without changing the acceptance condition at the largest accepted demand. Their signals preserve payoffs and unconditional military outcomes, although measures conditional on compliance can change. The reported coercive ability is for the pooling equilibrium.

The target's acceptance decision determines the pooling demand. Concession prevents war among types whose war costs fall between the two cutoffs. The target weighs this protection against the concession it gives up when war is avoided. It accepts as long as the expected war loss prevented is at least as large as that expected concession. At the largest accepted demand, $x^{\dagger}$, the target is indifferent.

PROPOSITION 1 (Benchmark Equilibrium and Coercive Ability): Under [Assumption 1](#) and interior war cutoffs at every accepted demand, pooling at $x^\dagger$ satisfies the intuitive criterion and attains the largest demand sustainable under pure target responses. At that demand, type payoffs and unconditional military outcomes are unique. After compliance, the coercer fights when $c_1 \leq \kappa^\dagger(x^\dagger)$, so settlement has probability $\alpha^\dagger = 1 - F(\kappa^\dagger(x^\dagger))$. Coercive ability is given by [\(2\)](#).

The acceptance inequality and the formula for $x^\dagger$ are derived in [SI §C.4. Proof of Propositions 1 and 2](#).

Proposition 1 determines both the demand and continuation behavior. Types below the cutoff fight after concession, while types above it settle. Holding the demand fixed, greater military power raises the cutoff and makes war optimal for a larger set of coercer types. The equilibrium demand also rises with power, so the equilibrium war probability need not rise.

Rearranging the target’s binding acceptance condition at $x^\dagger$ and substituting Definition 1 gives the subtraction form

$$\Omega^\dagger = \underbrace{(p + c_2) F(p + a)}_{\text{punishment after resistance}} - \underbrace{(p + c_2) F(\kappa^\dagger(x^\dagger))}_{\text{harm not prevented by compliance}} \quad (2)$$

Equation [\(2\)](#) writes coercive ability as one difference. The first term is the expected punishment produced by resistance. The second is the expected harm that compliance fails to prevent. A larger concession lowers the post-compliance war cutoff, makes settlement more likely, and reduces the second term. Greater power raises the first term and raises coercive ability as a whole, so whatever protection compliance loses is outweighed. The benchmark thus reproduces the expectation that power strengthens the threat faster than it erodes assurance ([Cebul, Dafoe, and Monteiro 2021](#)). The next section asks whether the subwar capability behaves the same way.

ADDING SUBWAR CAPABILITY

Now set $z > 0$ and add the subwar option. War costs rise faster with c_1 than operating costs, while settlement involves neither cost. When all three actions are optimal for some types, low-cost types fight, intermediate types use the operation, and high-cost types settle. Write $\kappa_{W(y)}$ for the boundary between war and the operation, and $\kappa_{L(y,r)}$ for the boundary between the operation and settlement. If these boundaries are reversed, the operation is dominated and the coercer chooses between war and settlement. SI [Lemma 3](#) gives the cutoff formulas.

Let W be the probability of war and L the probability of either war or the operation, evaluated under the prior. Thus $L - W$ is the probability of the operation and $1 - L$ the probability of settlement. Subscript R denotes resistance; $W(x)$ and $L(x)$ refer to compliance with demand x . The notation covers both an active and a dominated operation. If a demand changes beliefs, these probabilities use the target's posterior instead.

Before solving for the demand, hold transfer y fixed and compare threat and assurance. This exercise does not assume that the equilibrium demands are equal. Some types that would fight without the option instead use the operation. When war is more harmful, this substitution weakens the threat after resistance and strengthens assurance after compliance. Other types replace settlement with the operation. That substitution strengthens the threat after resistance and weakens assurance after compliance. The action replaced, not the operation alone, therefore determines how the target evaluates resistance and compliance.

Greater reach or lower relative cost makes the operation more attractive after either response. A larger audience cost also favors the operation after resistance because using it avoids the cost of backing down. A larger concession favors the operation over war after compliance because the operation preserves the transfer and war does not.

Having solved the coercer's continuation choice, turn to the target's response. The target compares the expected punishment from resistance with the expected cost of compliance. Compliance can cost it both the concession and any harm from force that follows.

It accepts when that combined cost is no greater than the punishment from resistance. The largest accepted demand makes the target indifferent. SI [Lemma 4](#) states the full acceptance condition.

Let α be the probability of settlement after compliance in the pooling equilibrium. A closed-form solution requires three additional conditions. The war cutoffs and the post-compliance operation–settlement cutoff are interior. The operation is active after compliance. Every nonfighting type uses the operation after resistance, so $\kappa_{L,R} \geq \bar{c}$ or $z + a \geq \lambda\bar{c}$. The interior post-compliance operation–settlement cutoff also implies $\alpha > 0$.

Each condition has a separate role. Interior cutoffs make action probabilities linear under the uniform distribution. Activity after compliance ensures that all three continuation actions matter. The condition on $\kappa_{L,R}$ sets $L_R = 1$, so every type that does not fight after resistance uses the operation. A positive α leaves some types that use the operation after resistance willing to settle after compliance. Compliance protects the target from the operation for exactly these types. [Assumption 1\(i\)](#) keeps $S(y) = z$ at accepted demands. These restrictions produce the closed-form solution. The substitution argument above does not require them.

PROPOSITION 2 (Equilibrium with a Subwar Operation): Under [Assumption 1](#) and the additional conditions above, a pooling equilibrium exists in which every type demands x^* and war occurs with probability $W(x^*)$. This equilibrium attains the largest demand sustainable under pure target responses and satisfies the intuitive criterion. At that demand, type payoffs and unconditional military outcomes are unique.

The formula for x^* is derived in SI [§C.4. Proof of Propositions 1 and 2](#), with the restrictions collected in SI [§C.2. The Closed-Form Region](#). On this branch, some types use force after resistance but settle after compliance, giving the target a reason to concede. If $\alpha = 0$, compliance provides no protection from the operation and the target accepts no positive

demand. The next result determines α and explains why a larger concession may fail to increase it.

Consider the operation–settlement choice after compliance. In the benchmark, a larger transfer raises the settlement payoff but leaves the war payoff unchanged. It therefore makes settlement more likely. With the operation, [Assumption 1\(i\)](#) gives $S(y) = z$ at every accepted demand. Settlement pays y , while the operation pays $y + z - \lambda c_1$, so their difference does not depend on the transfer.

PROPOSITION 3 (Settlement After Compliance): At a common accepted demand evaluated under the prior, if the operation is active and the operation–settlement cutoff is interior, [Assumption 1\(i\)](#) implies $\kappa_L(y) = \frac{z}{\lambda}$. The cutoff is constant in the demand, and $\alpha = 1 - F(\frac{z}{\lambda})$. Settlement becomes less likely as z rises and more likely as λ rises.

Proposition 3 characterizes one continuation boundary at an accepted demand, not the full acceptance decision. A larger transfer can move types from war to the operation through κ_W . It cannot move them from the operation to settlement because κ_L is constant. The largest accepted demand still depends on the target’s comparison of all expected costs after resistance and compliance. In [Figure 4](#), the constant cutoff appears as the horizontal upper boundary of the region where the operation replaces settlement. With positive escalation risk, this cutoff varies with the concession; [SI §F.1. Escalation Risk](#) gives the modified condition.

Now return to resistance. Whenever the operation is active after resistance, some types use it instead of war. Its availability therefore weakly reduces war after resistance. This comparison holds the target’s response fixed; it does not yet account for a change in the concession.

The probabilities W_R and α describe only two continuation actions. The resistance-stage comparison holds the history fixed and does not determine the equilibrium probability of war after the target complies, because the subwar capability can change the accepted

demand. To evaluate the target's full decision, convert every continuation outcome into expected loss. Let Θ^* denote punishment after resistance and Δ^* denote harm not prevented by compliance.

REMARK 1 (Coercive Ability as Threat Minus Assurance Failure): If a common accepted demand leaves the target indifferent, coercive ability equals expected punishment after resistance, Θ^* , minus harm not prevented by compliance, Δ^* . It is zero exactly when $\Theta^* = \Delta^*$. This identity follows by rearranging the target's binding acceptance condition and does not require the closed-form equilibrium.

$$\Omega^* = \underbrace{(p + c_2)W_R + z(L_R - W_R)}_{\Theta^*: \text{punishment after resistance}} - \underbrace{[(p + c_2)W(x^*) + S(x^*)(L(x^*) - W(x^*))]}_{\Delta^*: \text{harm not prevented by compliance}} \quad (3)$$

Each component adds expected war harm and expected subwar harm, evaluated after resistance for Θ^* and after compliance for Δ^* . War causes loss $p + c_2$; the operation causes loss z after resistance and $S(x^*)$ after compliance. The differences $L_R - W_R$ and $L(x^*) - W(x^*)$ are the corresponding probabilities of the operation alone.

Equation (3) is obtained by rearranging the target's binding acceptance condition: the expected concession it is willing to give up equals the punishment it avoids, less the harm it still expects after compliance. SI §D.1. [Proof of Remark 1](#) gives the derivation.

Let $\Theta^\dagger$ and $\Delta^\dagger$ be the corresponding benchmark quantities in (2). The operation increases coercive ability when its effect on punishment exceeds its effect on harm after compliance, and reduces coercive ability when the reverse holds. It weakens the threat by lowering Θ^* and weakens assurance by raising Δ^* . Which effect occurs depends on whether the operation replaces war or settlement after each target response. Removing the operation from both continuation menus recovers (2). Setting its gain to zero alone need

not reproduce the benchmark, because an operation can still avoid the audience cost after resistance.

WHEN THREAT AND ASSURANCE WEAKEN TOGETHER

To isolate the continuation choices, first hold transfer y fixed and compare harm excluding the retained concession, the component measured by Θ and Δ . Types that replace war with the operation change this harm from $p + c_2$ to z ; types that replace settlement with it add harm z . At resistance, where $y = 0$, this continuation-harm component equals the target's total loss.

With interior cutoffs and a uniform cost distribution, these two groups have a simple relation: for every type shifted out of war, $\frac{1-\lambda}{\lambda}$ types are shifted out of settlement. This continuation-harm component therefore rises when $z > \lambda(p + c_2)$, because the harm added by former settlers dominates, and falls when $z < \lambda(p + c_2)$, because replacing war dominates. The equality $z = \lambda(p + c_2)$ marks the balance between the two groups. SI §D.10. [Interior Switch Accounting](#) gives the cutoff algebra.

The same logic applies after resistance and compliance, but the number of types switching at each history differs. The audience cost makes the operation more attractive after resistance because it avoids the penalty for backing down. The accepted transfer makes it more attractive after compliance because the coercer keeps what the target has already paid. The net effect on coercive ability depends on how these different groups change punishment and post-compliance harm.

Two equilibrium responses qualify this simple comparison. One switching group may be exhausted. In the closed-form region, every type that avoids war after resistance already uses the operation, whereas some types still settle after compliance. The target can also respond by accepting a smaller demand, which may produce more war after compliance. Threat and assurance then weaken together when the loss of punishment from replacing war exceeds the harm added after resistance, while direct operation harm and the concession

response raise harm after compliance. Proposition 4 states the exact joint comparison, with its cutoff conditions in SI §D.10. [Interior Switch Accounting](#).

The link between the two components runs in one direction. Punishment after resistance is evaluated before any concession, so it does not vary with the demand. A weaker threat, however, lowers the largest accepted demand. The smaller concession makes war after compliance more attractive and raises the harm that compliance fails to prevent. This additional harm further reduces what the target will concede.

To isolate this feedback, hold the post-compliance continuation rules fixed as functions of the demand. In the closed-form region, the concession falls by more than the initial punishment loss when $x^* < p + c_2 - z$, meaning war is more costly for the target than conceding and then facing the operation. SI §D.1. [Proof of Remark 1](#) derives this response. Changing an operation parameter can also change the continuation rules, so this comparison isolates the effect of punishment.

Coercive Ability

When punishment falls and post-compliance harm rises, both changes reduce the concession that survives without war. More generally, coercive ability falls whenever the change in post-compliance harm exceeds the change in punishment, as (3) implies. Joint deterioration is therefore sufficient, although a large assurance loss can also outweigh a stronger threat.

PROPOSITION 4 (Joint Deterioration and Coercive Ability): Suppose the benchmark war cutoffs are interior at every accepted demand and the conditions for Proposition 2 hold. Threat and assurance both strictly weaken if and only if the two inequalities in SI [the joint-loss condition](#) hold. In that case, coercive ability is strictly lower. More generally, coercive ability falls exactly when the change in harm after compliance exceeds the change in punishment after resistance.

In the comparisons illustrated in [Figure 2](#), coercive ability rises more slowly with the probability of victory when the subwar capability is available. Added harm after compliance

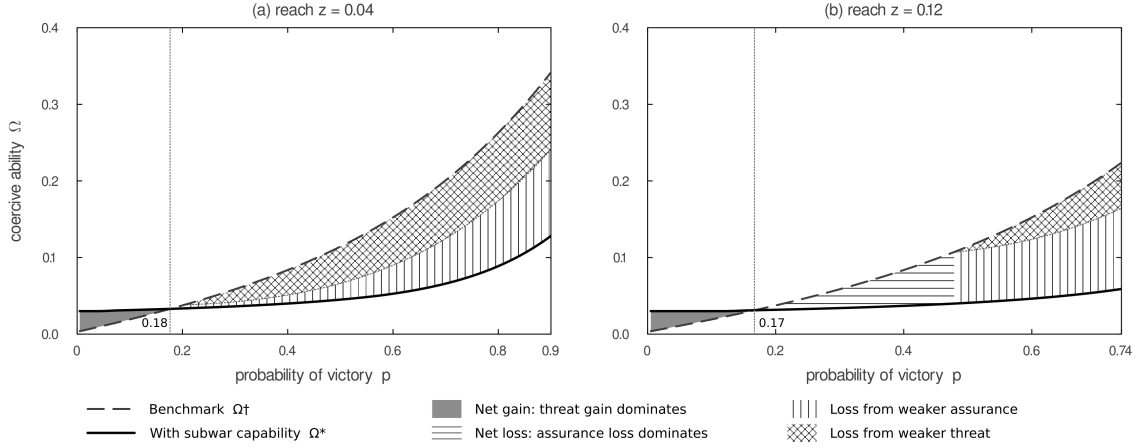


Figure 2: Illustration: Assurance Problems Weaken the Effect of Power on Coercive Ability

Note: Coercive ability in the benchmark (dashed) and with the subwar capability (solid) as the coerced’s probability of victory varies. The shaded gap is the difference between the two curves. Grey shading marks a net gain when the gain in threat outweighs the loss of assurance; horizontal lines mark a net loss when the assurance loss outweighs the threat gain. Where both weaken, vertical lines and crossed diagonal lines divide the loss into weaker assurance and weaker threat, respectively. Parameters: $c_2 = 0.02$, $a = 0.30$, $\lambda = 0.08$, c_1 uniform on $[0, 2]$; $z = 0.04$ in panel (a) and $z = 0.12$ in panel (b).

limits what the target will concede, so greater power produces smaller gains in coercive ability. The curves cross while punishment after resistance is still greater with the capability: assurance failure first outweighs the stronger threat, and weaker punishment later compounds the loss.

The two panels also differ in what produces the loss. At the smaller reach, no coerced type eventually chooses the operation after compliance; beyond that point the loss comes from weaker punishment after resistance together with the war that the smaller demand no longer prevents. At the larger reach, post-compliance harm combines the operation’s direct effect with additional war from the smaller demand. Proposition 4 gives the exact comparison. The panels illustrate its channels but do not establish their prevalence.

The exact comparison, including cases where threat and assurance move in opposite directions, is derived in SI §D.8. [Proof of Proposition 4](#). Its mechanism can be seen by separating the protection that compliance buys in the closed-form region:

$$\Omega^* = \underbrace{(p + c_2 - z)(W_R - W(x^*))}_{\text{protection from war}} + \underbrace{z\alpha}_{\text{protection from the operation}} \quad (4)$$

The first term values the reduction in war probability by the harm war adds beyond the operation. The second values the chance of settlement, which avoids the operation's harm as well. These two forms of protection determine how much the target will concede.

Greater reach makes avoiding the operation more valuable, but it also makes the coercer less willing to settle after compliance. The second term can therefore first rise and then fall as reach grows. Because reach changes both forms of protection, its overall effect on coercive ability cannot be inferred from the operation's harm or the war probability alone.

REMARK 2 (Power): In the closed-form region, the protection term $z\alpha$ depends on reach, relative cost, and the spread of war costs but not on the probability of victory. Total Ω^* still depends on power through both x^* and $p + c_2 - z$, so the derivative comparison with $\Omega^\dagger$ is not signed in general.

The operation–settlement cutoff identifies complete coercive failure.

REMARK 3 (Complete Coercive Failure): Under Assumption 1, coercive ability in the selected pooling equilibrium is zero if and only if $z \geq \lambda\bar{c}$. When the inequality holds, the operation–settlement cutoff after compliance is at or above the top of the type distribution, so even the highest-cost coercer type weakly prefers the operation to settlement after compliance and chooses it at a tie. Compliance then provides no protection from the operation, $\alpha = 0$, the target refuses every positive demand, and $x^* = \Omega^* = 0$.

The expression $\alpha = 1 - \frac{z}{\lambda\bar{c}}$ applies on the active branch only for $z \leq \lambda\bar{c}$; above that value the cutoff leaves the support and $\alpha = 0$. Remark 3 does not depend on the closed-form quadratic. If $z < \lambda\bar{c}$, a positive measure of types uses force after resistance but settles after compliance with zero, so the target strictly gains from a sufficiently small concession. If

$z \geq \lambda \bar{c}$, neither response to zero produces settlement, the target is indifferent there, and the strictly decreasing acceptance margin rules out every positive demand. The Supporting Information proves both claims.

The Coercer's Payoff

The concession response connects the coercer's private loss to the prospect of war. The same threshold determines when every coercer type is weakly worse off and when war is weakly more likely, even outside the closed-form region. Let $b = p - x^\dagger$ be the benchmark war cutoff and $d = x^\dagger - x^*$ the concession lost with the operation. The benchmark type at b is indifferent between war and settlement. It pays λb to operate, so the operation can compensate it for at most $z - \lambda b$ of lost concession.

PROPOSITION 5 (Payoff Loss and War Share a Threshold): Suppose $S(x^*) = z$ and $b = p - x^\dagger \in (0, \bar{c})$. Every coercer type is weakly worse off with the operation if and only if (5) holds. The same condition holds if and only if war after compliance is weakly more likely. War is strictly more likely exactly when the inequality is strict. If the condition holds and $d > 0$, every benchmark settler strictly loses, expected joint conflict cost is strictly higher, and coercive ability is strictly lower. If $d = 0$ and the condition holds, payoffs, war, conflict cost, and coercive ability are unchanged.

$$\underbrace{d}_{\text{concession lost}} \geq \underbrace{\max\{0, z - \lambda b\}}_{\text{net gain at the war margin}} . \quad (5)$$

The benchmark type at the war margin has the most to gain from the operation relative to its benchmark payoff. Lower-cost types already prefer war; higher-cost types pay more to operate. Once the lost concession exhausts that marginal type's net gain, no type benefits from the expanded menu. The same comparison determines whether the operation still prevents war among benchmark fighters. At the threshold, war is unchanged; above it, some benchmark settlers fight. The equivalence applies whether the operation remains active

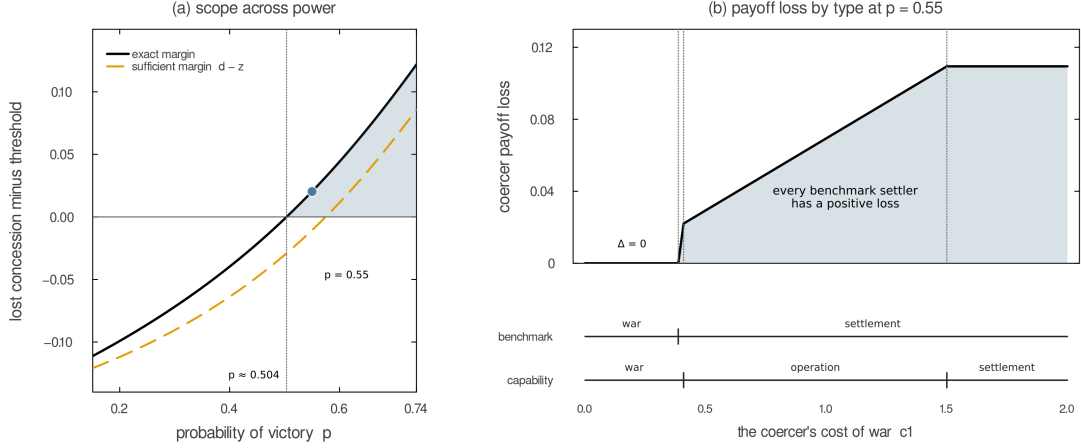


Figure 3: The Common Threshold for Payoff Loss and War

Note: Panel (a) plots the exact margin $x^\dagger - x^* - \max\{0, z - \lambda(p - x^\dagger)\}$ as a solid curve and the stronger sufficient margin $x^\dagger - x^* - z$ as a dashed guide. The exact margin crosses zero at $p \approx 0.504$, while the sufficient margin crosses at $p \approx 0.575$. Panel (b) fixes $p = 0.55$ and plots the benchmark payoff minus the operation-game payoff by coercer type; the aligned intervals show each game's post-compliance action. Parameters: $c_2 = 0.02$, $a = 0.30$, $\lambda = 0.08$, $z = 0.12$, and c_1 uniform on $[0, 2]$.

after compliance or is dominated, and does not require the closed-form equilibrium. When concessions shift military power or remain partly secure during war, the payoff-loss and war thresholds generally differ; SI §F.2. [Power Shift and Partial Recontestation](#) gives the separate conditions.

Figure 3 shows why accounting for the operation's cost matters. At the illustrated $p = 0.55$, the lost concession is smaller than the operation's gross gain but exceeds its net gain for the benchmark type at the war margin. Every benchmark settler therefore loses, and war rises. Types that fight in both games are unaffected. The payoff loss can come from switching from settlement to war, paying to operate, or settling for a smaller concession.

Proposition 6 separates the two effects on the coercer's expected payoff: gaining another military option at a fixed concession and receiving a different concession from the target.

PROPOSITION 6 (The Coercer's Expected Payoff): Suppose the relevant cutoffs are interior and the operation is active after concession. The change in expected payoff equals the gain from adding the operation at the new concession plus the

payoff effect of changing the concession in the benchmark. The first component is nonnegative. The second is negative when $x^* < x^\dagger$. Expected payoff falls exactly when the payoff loss from the smaller concession outweighs the gain from the additional option.

Holding the concession fixed, the operation cannot hurt the coercer because every original action remains available. A smaller accepted concession can nevertheless make possession of the capability costly overall. SI §D.4. [Proof of Proposition 6](#) gives the payoff formulas and the exact decomposition.

SUBWAR CAPABILITY AND THE PROSPECT OF PEACE

The operation changes both the available continuation actions and the concession at which the coercer chooses among them. At a fixed concession, it can replace war among low-cost types or settlement among intermediate types below κ_L . When the accepted demand falls, however, some benchmark settlers choose war. These types never use the operation. [Figure 4](#) compares the resulting post-compliance actions across the two games.

Proposition 5's threshold applies whether the operation remains active after compliance or becomes dominated. War must be preferable to both alternatives. While the operation is active, it sets the relevant war cutoff; otherwise settlement does. In either case, war rises when the smaller concession makes some benchmark settlers prefer to fight.

Panel (b) distinguishes two ranges in which war exceeds its benchmark probability. Between the crossing at $p \approx 0.29$ and the dotted line at $p \approx 0.57$, the operation remains active after compliance. Its anticipated use weakens assurance directly. In the hatched range above $p \approx 0.57$, the operation is dominated after compliance and is never used on the equilibrium path. It nevertheless replaces war after resistance, weakening punishment and reducing the concession. In both ranges, the smaller concession draws benchmark settlers into war and thereby weakens assurance further. These effects are jointly determined in

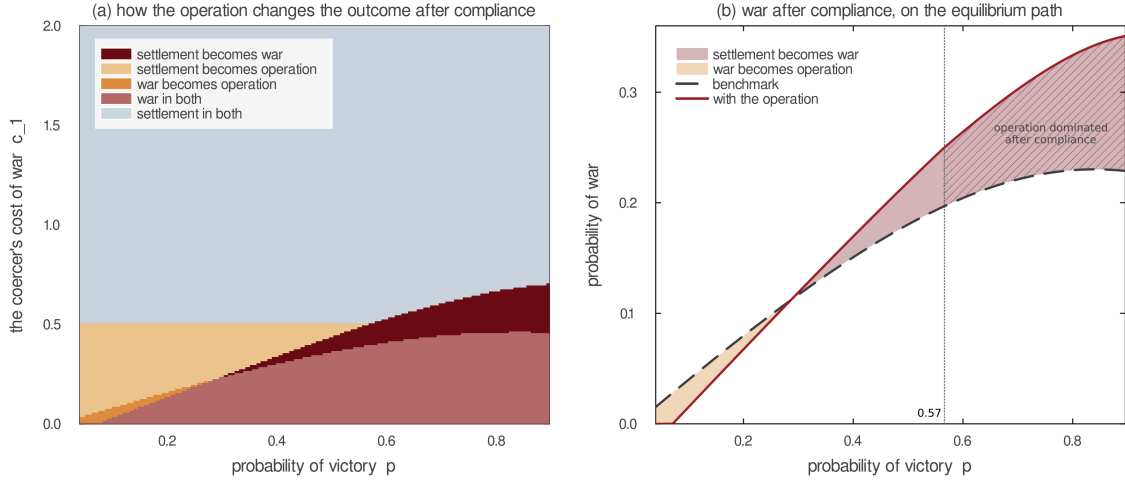


Figure 4: Subwar Capability Can Replace War Yet Increase It

Note: Panel (a) colors each combination of the coercer’s probability of victory and war cost by the change in the post-compliance outcome when the operation becomes available, at the calibration of Figure 2 with $z = 0.04$. The operation region’s upper boundary is horizontal because $\kappa_L = \frac{z}{\lambda}$ contains neither p nor the demand. Panel (b) plots the probability of war after compliance on each game’s equilibrium path: below $p \approx 0.29$ the operation replaces war; above it, the smaller accepted demand pushes types that would have kept the benchmark settlement into war. Hatching marks the range above $p \approx 0.57$, where the operation is dominated after compliance. There the post-compliance menu is the benchmark menu evaluated at a smaller demand, and the entire increase follows from the weaker punishment the target expects after resistance.

equilibrium, not a temporal sequence. Additional war consists of direct choices of war by types that never use the operation.

The unused-operation range has a distinct implication for studying capability. Possession can affect bargaining through the punishment expected after resistance even when no operation is observed after compliance. An analysis based only on observed operations would miss that channel. Assessing it would require information about capability possession, the demand and concession, the target’s response, and subsequent violence, including crises in which the operation is never used. The model supplies this distinction between possession and use; establishing its causal importance requires a research design that addresses which states possess the capability and which crises enter the comparison.

Subwar Capability and the Cost of Conflict

The subwar capability changes the expected joint cost of conflict, not only the probability of war. Let K denote expected joint conflict cost in the model. Settlement costs nothing, war costs the two states $c_1 + c_2$, and the operation costs λc_1 . The baseline treats both the demand and the operation's gain as transfers between the states, so modeled joint welfare is $1 - K$. This accounting excludes any physical destruction whose loss to the target is not offset by the coercer's modeled gain.

Concession loss. When the operation initially saves costs, that saving can disappear before war returns to its benchmark rate. To isolate this ordering, hold the operation's primitives fixed and vary the concession lost. Let $\Delta K(d)$ be operation-game conflict cost after concession $x^\dagger - d$ minus benchmark conflict cost. Thus $\Delta K(x^\dagger - x^*) = K^* - K^\dagger$, while $\Delta K(0)$ is the direct cost effect at the benchmark concession. Along $0 \leq d \leq x^\dagger$, require an interior benchmark war cutoff and interior relevant continuation cutoffs at every evaluated transfer, including $d = 0$. The operation may leave the menu, provided the remaining war-settlement cutoff stays interior. Suppose the operation reaches some benchmark settlers, $\frac{z}{\lambda} > b$, and define the war threshold $d_W = z - \lambda b$ with $0 < d_W < x^\dagger$.

Under these path conditions, Proposition 7 establishes that conflict cost strictly increases with the lost concession. While the operation is active, a smaller concession moves types from the operation into war; once it is dominated, a further reduction moves types from settlement into war. If $\Delta K(0) < 0$, the initial saving is exhausted at a unique $d_K \in (0, d_W)$. Cost rises above its benchmark level before war does because, when war returns to its benchmark probability, the operation still replaces settlement for some types and adds λc_1 in costs.

Lower operating cost. Making the operation cheaper changes both its use and the concession. These comparisons are local and do not require the whole lost-concession path to have interior cutoffs. Hold p, c_2, a, z , and $\bar{c}$ fixed on an active interior post-compliance branch. At a fixed concession, lower operating cost saves resources on operations already

conducted and encourages additional use. Operations replacing war can save costs; operations replacing settlement add costs. Total conflict cost rises when those added costs outweigh the savings from cheaper operations and avoided war. SI §D.5. Proof of Proposition 7 gives the exact condition.

When the concession adjusts, maintain Proposition 2's closed-form conditions with $z + a > \lambda \bar{c}$, so no type settles after resistance and this restriction is strict.

PROPOSITION 7 (Concession Loss and Operating Cost): *Concession loss.* Under the stated path conditions, $\Delta K(d)$ strictly increases and, if $\Delta K(0) < 0$, crosses zero at a unique $d_K \in (0, d_W)$.

Lower operating cost. In the stated closed-form region, cheaper operations reduce the accepted demand. They raise equilibrium conflict cost whenever cheaper operations already raise cost at the fixed equilibrium concession. This sufficient condition always holds when $c_2 = 0$.

After resistance, a cheaper operation replaces war and reduces punishment. After compliance, it can replace war or settlement, with opposing effects on protection. Accounting for both substitutions, the positive demand derivative establishes that the target concedes less as λ falls. That response moves additional types toward war. Throughout Figure 5's range, the settlement substitution also dominates the direct post-compliance comparison, so both punishment and assurance weaken.

The smaller equilibrium concession raises cost further. Thus, whenever the added operations outweigh the direct cost savings at a fixed concession, the bargaining response compounds the increase. As λ falls from 0.21 to 0.061 in Figure 5, cost at the benchmark concession rises from 0.051 to 0.087. Allowing the concession to adjust raises equilibrium cost from 0.057 to 0.138, while benchmark cost remains 0.050.

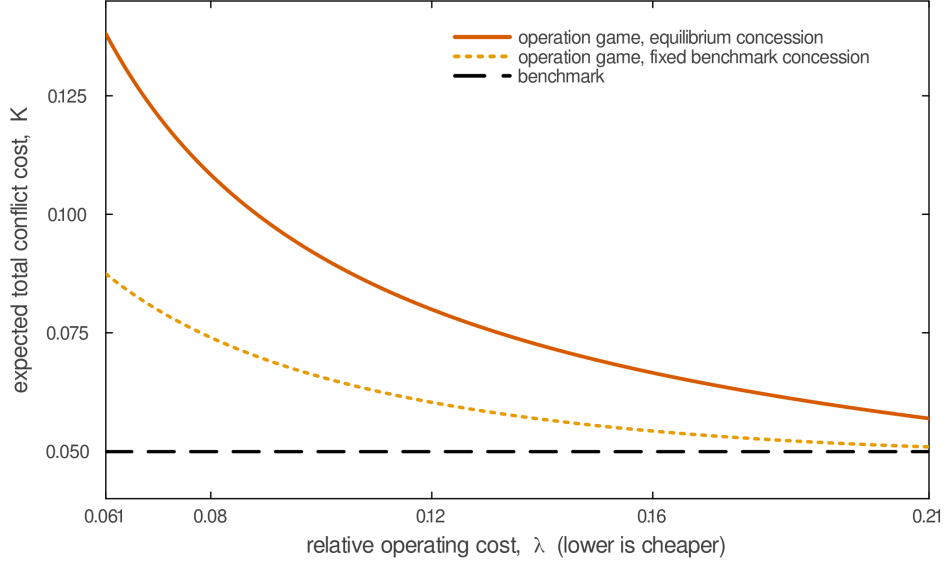


Figure 5: Cheaper Operations Can Raise Total Conflict Cost

Note: The solid curve plots equilibrium expected total conflict cost with the operation. The dotted orange curve holds the concession at its benchmark level, isolating the direct effect of changing λ . The dashed black line is benchmark cost. Lower λ means a cheaper operation. Over the displayed interval, lowering λ weakens the threat, weakens assurance on net, and reduces the accepted concession. The interval satisfies the closed-form premises and the fixed-concession cost condition in SI §D.5. [Proof of Proposition 7](#) Proposition 7 gives the positive demand derivative. The upper endpoint $\lambda = 0.21$ satisfies $z + a = \lambda \bar{c}$, so its derivative comparison is from below. Parameters: $p = 0.65$, $c_2 = 0.02$, $a = 0.30$, $z = 0.12$, and c_1 uniform on $[0, 2]$.

OPERATIONAL GAIN AND POLITICAL COST

In the baseline, z is at once the coercer's gain and the target's harm. To separate their effects, let an operation impose harm $\eta > 0$ on the target and deliver gain $g \geq 0$ to the coercer, with the two unlinked. Treat g and η as fixed throughout the extension's feasible demand set, and suppose the relevant post-resistance cutoffs are interior and the selected demands are below the upper demand bound.

COROLLARY 1 (Gain and Harm): At a common conceded demand x evaluated under the prior, settlement occurs with probability $\alpha = 1 - F(\max\{p - x, \frac{g}{\lambda}\})$. On the active operation branch this is $1 - F(\frac{g}{\lambda})$, independent of the target's harm. If $g = 0$ and the operation is active after resistance, the subwar capability raises the accepted demand and coercive ability if and only if $\eta > \lambda(p + c_2)$.

Settlement requires both war and the operation to be less attractive than accepting the concession alone; the larger of their two cost cutoffs therefore determines its probability. The target's loss η governs how much harm each operation does. The coercer's gain g governs whether the operation itself remains attractive after compliance. An operation that provides no benefit still costs λc_1 to conduct, so no positive-cost type conducts it once the transfer is in hand. This direct exposure to the operation depends on its benefit to the coercer. War can still follow compliance, and the demand response can change its probability even when $g = 0$.

An operation with audience cost $a_L \in [0, a]$ becomes less attractive after both responses. The cost of backing down, a , instead makes the operation relatively attractive after resistance. SI §F.3. [The Operation's Audience Cost](#) gives the resulting cutoffs and acceptance equation, including the conditions under which the operation replaces war or leaves the continuation menu.

The bound on operational gain differs from issue indivisibility. The stake remains divisible here; the bound limits what the coercer can obtain through the operation after the target responds. [Spaniel \(2023\)](#) instead restricts the division of the stake and shows that moderate indivisibility can promote peace by constraining later demands or inducing safer offers.

SCOPE OF THE EQUILIBRIUM COMPARISON

Part of the analysis does not depend on equilibrium selection. If the operation takes a fixed amount and is active after an accepted demand, the coercer settles if and only if $c_1 > \frac{z}{\lambda}$. That cutoff does not depend on the demand or off-path beliefs, and SI [Proposition 8](#) proves it for every equilibrium. The threat-minus-assurance identity in Remark 1 requires only a common accepted demand that leaves the target indifferent. The cutoff result additionally requires the operation to be active at that demand. Different equilibria can place different demands on the path and associate different types with them, so the cutoff alone does not determine the unconditional settlement rate or coercive ability.

Pure target responses admit an interval of largest accepted demands. SI [Lemma 5](#) characterizes this interval under Assumption 1 and shows that every demand in it supports a pooling equilibrium satisfying the intuitive criterion. Types that strictly benefit from compliance choose the largest accepted demand; indifferent types may send other signals. At a fixed largest demand, those signals preserve type payoffs and unconditional military outcomes.^[8]

The selected equilibrium maximizes every coercer type's payoff and the unconditional retained concession. It also minimizes unconditional war and expected joint conflict cost within the pure-response class, as SI [Corollary 2](#) shows. A larger concession raises the payoffs from settlement and the operation relative to war. Thus, when the capability raises war or conflict cost in the selected comparison, it raises the corresponding minimum across equilibria. The comparison does not rank every pair of equilibria across the games or establish which equilibrium an actual crisis selects.

Zero demand can coexist with positive accepted demands. In the benchmark, it is an equilibrium exactly when $p + a < \bar{c}$. In the operation game's region without settlement

^[8]They can change the probability of compliance and the posterior at an accepted demand. War and retained concessions conditional on compliance can therefore differ across signal strategies. The reported coercive-ability formulas describe the selected pooling assessments.

after resistance, it is an equilibrium exactly when $z > \lambda p$. If $z \geq \lambda \bar{c}$, every pure-response equilibrium instead yields zero retained concession.^[9]

The intuitive criterion permits this multiplicity and does not select the largest demand. Under the interior-war conditions of Propositions 1 and 2, both selected positive outcomes fail D1, although some zero-demand outcomes in the operation game survive it.^[10]

CONCLUSION

A subwar capability can undermine the bargain that makes restraint worthwhile. A subwar operation can weaken punishment after resistance and protection after compliance, reducing the target's concession. The smaller concession makes war more attractive after compliance. Under the baseline conditions, the same threshold determines when every coercer type is weakly worse off and when war is weakly more likely.

These results change what observed force can reveal about capability. An operation that goes unused can still weaken the threat after resistance, reduce the concession, and induce war among benchmark settlers. Conversely, fewer general wars need not imply lower modeled conflict costs: when an initial operating-cost saving exists under the concession-path conditions, it disappears before war returns to its benchmark rate. Studying capability therefore requires attention to both the bargains it changes and the forms of force that follow them.

These results suggest why a subwar capability's value in deterrence need not carry over to compellence. Both require punishment to be conditional, but the compellence problem here asks the target to make a concession and then rely on the coercer not to use the

^[9]SI §E.1. [Zero-Demand Equilibria](#) gives the full conditions, including settlement after resistance and cases in which zero is the only attainable demand.

^[10]D1 compares types' incentives to deviate when restricting off-path beliefs. SI [Proposition 9](#) shows why positive outcomes with an interior war cutoff fail: concession can move resolute types away from war, making them favorable news for the target. If $c_2 > 0$, the elimination is independent of the tie rule; at $c_2 = 0$, it is tie-breaking-sensitive. The analysis imposes the intuitive criterion and does not claim refinement invariance.

operation. Because the operation remains available after compliance, it can strengthen punishment after resistance while weakening the protection that compliance buys. A subwar option may therefore support a deterrent threat yet weaken a compellent bargain. The effect of a subwar capability on compellence cannot be inferred from threat credibility alone (Schelling 1966; Kydd and McManus 2017; Cebul, Dafoe, and Monteiro 2021).

The analysis also qualifies the usual option-value logic. At a fixed transfer, adding the operation cannot hurt the coercer because every benchmark action remains available. When the coercer makes the largest demand the target will accept, however, the expanded menu can reduce the transfer and make the coercer worse off. This suggests a role for commitments that preserve punishment after resistance while limiting the operation after compliance. Such a restriction could improve the earlier bargain by protecting the target after it concedes, an application of the power to bind oneself (Schelling 1960).

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SUPPORTING INFORMATION

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A. NOTATION

Histories and payoffs. Let $(y, r) = (x, 0)$ after concession and $(y, r) = (0, 1)$ after resistance. A subwar operation produces a gain $S(y) = \min\{z, 1 - y\}$ and costs the coercer λc_1 , so $S(0) = z$. Under [Assumption 1\(i\)](#), $S(y) = z$ at every accepted demand. Settlement costs the coercer $a\mathbb{1}[r = 1]$, and the target's loss from war is $p + c_2$.

Continuation probabilities. Superscript R denotes the resistance history, $C(0)$ denotes compliance with zero, and an unmarked positive argument x denotes compliance. Under the prior, W is the probability of war and L is the probability of either war or the operation at the specified history. When the operation is active, $W = F(\kappa_W)$ and $L = F(\kappa_L)$; when it is dominated, $W = L = F(\kappa_S)$. At a demand that changes beliefs, these probabilities use the posterior instead. Finally, $\alpha = 1 - L(x^*)$ is the probability of settlement after compliance in the pooling equilibrium.

Comparisons between games. A dagger marks benchmark quantities; $x^\dagger$ and x^* are the selected demands in the benchmark and operation game. The benchmark marginal type is $b = p - x^\dagger$, and $d = x^\dagger - x^*$ is the lost concession in the payoff and conflict-cost comparisons. In [Corollary 1](#), η denotes the operation's harm to the target, while g denotes the coercer's gain.

Outcomes. Coercive ability Ω measures the accepted concession retained without general war. Expected punishment after resistance is Θ , and harm not prevented by compliance is Δ . The coercer's expected total payoff is V . Expected joint conflict cost is K , so modeled joint surplus is $1 - K$.

A.1. Results, Conditions, and Proofs

The table locates the conditions and proofs for the main comparisons. The linked statements give each result's full scope.

Result	Conditions	Proof or derivation
Proposition 1	Assumption 1 and interior benchmark war cutoffs.	§C.4. Selected equilibria
Proposition 2	Assumption 1; §C.2. Closed-Form Region .	§C.4. Selected equilibria §C.1. Off-path beliefs
Proposition 3	An uncapped accepted demand and an active operation with an interior settlement cutoff.	§D.2. Settlement after compliance
Corollary 1	Fixed gain and harm; the demand comparison additionally uses zero gain and an active operation with interior resistance cutoffs.	§D.3. Gain and harm
Remark 1	A common accepted demand leaving the target indifferent.	§D.1. Threat and assurance
Proposition 4	The benchmark and operation-game closed-form conditions.	§D.8. Coercive ability §D.10. Joint deterioration
Joint threat and assurance loss	Interior switching at fixed transfers; joint-loss conditions in the closed-form region.	§D.10. Interior switch accounting
Proposition 5	Full operation reach at the selected demand and an interior benchmark war cutoff.	§D.9. Payoff loss and war
Proposition 6	Interior cutoffs and an active operation after compliance.	§D.4. Expected payoff
Proposition 7	Separate conditions for the concession-loss path and the local relative-cost comparison.	§D.5. Joint conflict cost
War after resistance	Operation activity and the supported resistance cutoffs.	§D.7. Resistance-stage comparison
Proposition 8	An active operation of fixed gain at the conceded demand.	§D.6. Selection-free cutoff
Lemma 5	Assumption 1, pure target responses, and concession at indifference.	§E. Complete equilibrium interval
Corollary 2	The pure-response equilibrium class in Lemma 5.	Minimum war and conflict cost
Proposition 9	A positive largest accepted demand, pure target responses, and an interior war cutoff.	§E.4. D1 comparison §E.1. Zero-demand cases

The extensions derive their conditions alongside their results: [§F.1. escalation risk](#), [§F.2. power shift and partial recontestation](#), and [§F.3. the operation's audience cost](#).

B. PROOFS OF THE LEMMAS

LEMMA 1 (The Coercer's Menu in the Benchmark): Given history (y, r) , the coercer fights if and only if $c_1 \leq \kappa^\dagger(y, r) = p - y + a\mathbb{1}[r = 1]$ and settles otherwise.

LEMMA 2 (No Type Prefers Resistance): For every c_1 and $x > 0$, the coercer's continuation value is weakly higher after concession than after resistance and strictly higher for types that do not fight after concession. If the operation is active after concession, the indifferent types are $[0, \kappa_W(x)]$, which has positive measure if and only if $p > x + S(x)$. If the operation is dominated, the indifferent types are $[0, \kappa_S(x)]$.

LEMMA 3 (The Coercer's Menu with a Subwar Operation): If $\kappa_W(y) < \kappa_L(y)$, the coercer fights below $\kappa_W(y)$, uses the operation between the cutoffs, and settles above $\kappa_L(y)$. If the cutoffs are reversed, the operation is dominated. Equation (6) gives both cutoffs.

$$\kappa_W(y) = \frac{p - y - S(y)}{1 - \lambda} \quad \text{and} \quad \kappa_L(y, r) = \frac{S(y) + a\mathbb{1}[r = 1]}{\lambda} \quad (6)$$

At history (y, r) , an active operation gives $W(y) = F(\kappa_W(y))$ and $L(y) = F(\kappa_L(y, r))$. If the operation is dominated, define $\kappa_{S(y, r)} = p - y + a\mathbb{1}[r = 1]$ and set $W(y) = L(y) = F(\kappa_{S(y, r)})$. When $S(y) = z$, the operation is active after resistance if and only if $z + a > \lambda(p + a)$, and after concession of x if and only if $z > \lambda(p - x)$.

LEMMA 4 (The Target's Rule): Under the prior, the target concedes x if and only if (7) holds. For any other posterior, the same comparison uses continuation probabilities computed under that posterior.

$$\begin{aligned} & (p + c_2)W_R + z(L_R - W_R) \\ & \geq x(1 - W(x)) + (p + c_2)W(x) + S(x)(L(x) - W(x)) \end{aligned} \quad (7)$$

The left side is expected punishment after resistance. On the right, the first term is the expected retained concession; the remaining terms are harm that compliance does not prevent.

B.1. Proof of Lemma 1

Proof. At history (y, r) , settlement pays $y - a\mathbb{1}[r = 1]$ and war pays $p - c_1$. The coercer fights when $p - c_1 \geq y - a\mathbb{1}[r = 1]$. Rearranging this inequality gives the cutoff in the lemma. $\square$

B.2. Proof of Lemma 2

Proof. Fix c_1 and $x > 0$. Settlement pays x after concession and $-a$ after resistance. War pays $p - c_1$ after either response. The operation pays $x + S(x) - \lambda c_1$ after concession and $z - \lambda c_1$ after resistance. If $z \leq 1 - x$, then $x + S(x) = x + z > z$. If $z > 1 - x$, then $x + S(x) = 1 > z$. Thus each continuation action pays at least as much after concession as after resistance.

War is the only action whose payoff does not strictly rise after concession. Because the war cutoff weakly decreases with y , a type that fights after concession also fights after resistance. That type receives $p - c_1$ after either response and is indifferent.

Now take a type that does not fight after concession. Because war was available but not chosen, its continuation value after concession exceeds $p - c_1$. Settlement and the operation also pay strictly more after concession because $x > -a$ and $x + S(x) > z$. The type's continuation value after resistance is the maximum of $p - c_1$, $-a$, and $z - \lambda c_1$. Each candidate is strictly below its corresponding post-concession payoff or below the post-concession continuation value. The type therefore strictly prefers concession.

If the operation is active, the indifferent set is exactly $[0, \kappa_W(x)]$ and has positive measure if and only if $p > x + S(x)$. If the operation is dominated, the indifferent set is $[0, \kappa_S(x)]$. These types' payoffs do not determine their demands. The pooling equilibrium assigns them x^* .

No type gains by inducing resistance. Settling types prefer the largest accepted demand. Operation users weakly prefer a larger demand and strictly prefer it under Proposition 2's condition that $S(y) = z$. Fighting types are indifferent. The pooling equilibrium assigns the same demand to every type, so the target evaluates that demand under the prior F . $\square$

B.3. Proof of Lemma 3

Proof. Fix history (y, r) . The coercer's three payoffs are

$$u_S = y - a\mathbb{1}[r = 1], \quad u_W = p - c_1, \quad u_L = y + S(y) - \lambda c_1$$

Their slopes in c_1 are 0, -1 , and $-\lambda$. Since $-1 < -\lambda < 0$, the optimal action changes from war to the operation and then to settlement as c_1 rises. Pairwise indifference gives the cutoffs in (6): $u_W = u_L$ at κ_W , $u_L = u_S$ at κ_L , and $u_W = u_S$ at κ_S .

The operation exceeds both other payoffs on a nonempty interval of cost values if and only if $\kappa_W < \kappa_L$. A positive measure of supported types uses it exactly when $\max\{0, \kappa_W\} < \min\{\bar{c}, \kappa_L\}$. If $\kappa_W \geq \kappa_L$, its payoff never exceeds $\max\{u_W, u_S\}$. The coercer then fights below κ_S and settles above it. $\square$

B.4. Proof of Lemma 4

Proof. After concession, war occurs with probability $W(x)$ and gives the target $1 - p - c_2$. The operation occurs with probability $L(x) - W(x)$ and gives the target $1 - x - S(x)$. Settlement occurs otherwise and gives the target $1 - x$. Hence

$$V_2(x) = W(x)(1 - p - c_2) + (1 - x)(1 - W(x)) - S(x)(L(x) - W(x))$$

The target's payoff after resistance is $V_{2,R} = 1 - (p + c_2)W_R - z(L_R - W_R)$, evaluated at history $(0, 1)$. Compliance with zero instead uses history $(0, 0)$. The target concedes if and only if $V_2(x) \geq V_{2,R}$. Rearranging yields (7). $\square$

REMARK 4 (Distributional Scope): Let $h_F(c) = \frac{f(c)}{1-F(c)}$ be the upper-tail hazard of a differentiable war-cost distribution. On the uncapped domain, the sufficient condition

$$\sup_{0 \leq c \leq \frac{p}{1-\lambda}} h_F(c) < \frac{1-\lambda}{p+c_2}$$

makes the target's acceptance margin strictly decreasing, so its accepted demands form an interval. For the uniform distribution this condition is exactly [Assumption 1\(ii\)](#). The continuation cutoffs, the target rule, [Proposition 3](#), [Proposition 8](#), and the threat–assurance identity do not require uniformity. The quadratic demand formulas use uniformity; strict monotonicity and the resulting acceptance-interval conclusion use the stated hazard bound. Outside the hazard condition, the paper makes no connected-acceptance-set claim without a global solver.

Proof. Let $G(x) = V_2(x) - V_{2,R}$ be the target's acceptance margin. In a region where the operation is dominated after compliance,

$$G'(x) = f(p-x)(p+c_2-x) - [1-F(p-x)].$$

In an active, uncapped region,

$$G'(x) = f(\kappa_W(x)) \frac{p+c_2-x-z}{1-\lambda} - [1-F(\kappa_W(x))].$$

If either parenthesized harm term is nonpositive, the corresponding derivative is negative immediately. Otherwise divide by the positive survival probability. Since every relevant cutoff lies below $\frac{p}{1-\lambda}$ and each harm term is at most $p+c_2$, the stated upper-tail-hazard bound makes both derivatives strictly negative. The payoff, and hence G , is continuous where the operation enters or leaves the coercer's envelope, so the accepted set is an interval on the uncapped domain. For $F(c) = \frac{c}{\bar{c}}$, $h_F(c) = \frac{1}{\bar{c}-c}$; its supremum on the stated interval is $\frac{1}{[\bar{c}-\frac{p}{1-\lambda}]}$. Substitution reduces the bound to $\bar{c} > \frac{2p+c_2}{1-\lambda}$. $\square$

C. PROOF OF THE EQUILIBRIUM CHARACTERIZATION

C.1. Off-Path Beliefs and the Intuitive Criterion

The following construction applies to the benchmark, the operation game, and the extensions below. Let $U_{C(x, c_1)}$ and $U_{R(c_1)}$ denote the coercer's optimized continuation payoffs after concession of x and after resistance. Suppose $U_{C(x, c_1)}$ is weakly increasing in x for every type, x^e is the largest demand accepted under the prior, and $U_{R(c_1)} \leq U_{C(x^e, c_1)}$ for every type. Then pooling at x^e can be supported as a perfect Bayesian equilibrium satisfying the intuitive criterion. This statement permits $x^e = 0$ and does not require interior continuation cutoffs.

Proof. I construct beliefs and responses at every unused demand. On path, all types demand x^e , the posterior is the prior, and the target concedes. Continuation choices maximize the payoffs defining U_C and U_R . Write $u^{e(c_1)} = U_{C(x^e, c_1)}$.

For an unused demand x' , let $\mathcal{B}(x')$ contain the concession probabilities that are best responses to some posterior on the full type space. A type survives equilibrium dominance exactly when

$$u^{e(c_1)} \leq \max_{q \in \mathcal{B}(x')} \{qU_{C(x', c_1)} + (1 - q)U_{R(c_1)}\}.$$

Denote the surviving set by $\mathcal{T}(x')$. This is the continuous-type version of the intuitive criterion's first step (Cho and Kreps 1987). When this set is nonempty, I choose a posterior supported on it and a best response that deters every type. When it is empty, the criterion imposes no belief restriction. Every constructed response follows the maintained rule of concession at indifference.

Smaller demands. If $x' < x^e$, monotonicity and the resistance-payoff bound give

$$U_{C(x', c_1)} \leq u^{e(c_1)}, \quad U_{R(c_1)} \leq u^{e(c_1)}.$$

Hence no response, including a mixture, makes any type strictly better off. If $\mathcal{T}(x')$ is nonempty, choose any posterior supported on that set and its target best response. If it is empty, use the prior and its best response. Neither prescription creates a profitable deviation.

Larger demands. If $x' > x^e$ and concession is optimal under some posterior, then $1 \in \mathcal{B}(x')$. Monotonicity gives $U_{C(x', c_1)} \geq u^{e(c_1)}$ for every type, so every type survives. Use the prior:

the target strictly rejects x' because x^e is its largest accepted demand and equality requires concession. Rejection pays $U_{R(c_1)} \leq u^{e(c_1)}$ and deters every type.

If concession is optimal under no posterior, rejection is optimal under every posterior. The surviving types are exactly those with $U_{R(c_1)} = u^{e(c_1)}$. Use a posterior on that set if it is nonempty, and the prior otherwise. Rejection is sequentially rational and again deters every type. These cases exhaust all unused demands, including when $x^e = 0$. Together with Bayes' rule on path and optimal continuation choices, the construction proves the claimed refined pooling equilibrium. $\square$

C.2. The Closed-Form Region

Beyond [Assumption 1](#), Proposition 2's closed form requires

$$\lambda(p - x^*) < z < p - x^*, \quad \lambda\bar{c} - a \leq z < \lambda\bar{c}, \quad z \leq \frac{1 - p - c_2}{2}$$

The first pair makes the operation active after compliance and keeps the post-compliance war cutoff interior. The second pair combines $\kappa_{L,R} \geq \bar{c}$ with $\alpha > 0$ and implies

$$\alpha \leq \min\left\{1, \frac{a}{\lambda\bar{c}}\right\}$$

so the probability of settlement after compliance is capped strictly below one whenever $a < \lambda\bar{c}$. The condition $\kappa_{L,R} \geq \bar{c}$ also sets $L_R = 1$: no coercer type settles after resistance, so the post-resistance menu is effectively binary while the post-compliance menu is ternary. Results based on the operation-game quadratic therefore describe a region in which backing down never occurs, which is the asymmetry that makes Θ^* large and the truncation in [\(20\)](#) one-sided.

C.3. The Two Selected Equilibria

C.4. Proof of [Proposition 1](#) and [Proposition 2](#)

Proof. Benchmark. Let $G^\dagger(x)$ be the target's payoff from concession minus its payoff from resistance. The expected concession is no greater than the war loss avoided exactly when

$$x(1 - F(p - x)) \leq (p + c_2)[F(p + a) - F(p - x)]$$

With interior cutoffs and the uniform distribution, this condition becomes

$$x^2 + x(\bar{c} - 2p - c_2) - (p + c_2)a \leq 0$$

The constant term is negative, so the quadratic has one positive and one negative root. The target accepts every nonnegative demand up to the positive root and rejects every larger demand. The positive root is (8).

$$x^\dagger = \frac{1}{2} \left[-(\bar{c} - 2p - c_2) + \sqrt{(\bar{c} - 2p - c_2)^2 + 4(p + c_2)a} \right] \quad (8)$$

At this demand, war occurs with probability $F(p + a)$ after resistance and $F(\kappa^\dagger(x^\dagger))$ after concession. Definition 1 and the binding acceptance condition then give (2) and $\alpha^\dagger = 1 - F(\kappa^\dagger(x^\dagger))$.

Operation game. Let $G(x) = V_2(x) - V_{2,R}$. First consider demands for which the operation is dominated after concession. Then $W(x) = L(x) = F(p - x)$, and the uniform distribution gives

$$G'(x) = \frac{2p + c_2 - \bar{c} - 2x}{\bar{c}} < 0$$

Assumption 1(ii) implies $\bar{c} > 2p + c_2$, so this derivative is negative. Next consider demands for which the operation is active after concession and $S(x) = z$. Then $L(x) = F(\frac{z}{\lambda})$, and

$$G'(x) = \frac{f(\kappa_W)(p + c_2 - x - z)}{1 - \lambda} - (1 - F(\kappa_W))$$

This derivative is negative if $\frac{2p + c_2 - 2x - 2z}{1 - \lambda} < \bar{c}$. Its left side is no larger than $\frac{2p + c_2}{1 - \lambda}$, which is below $\bar{c}$ under the maintained restriction. The target's payoff is continuous where the operation enters the coercer's optimal menu. If the post-compliance war cutoff falls below zero, the corresponding derivative is -1 ; Assumption 1(ii) excludes a cutoff above the support. Thus G is strictly decreasing across the dominated and active regions wherever $S(x) = z$.

At compliance with zero, the audience cost does not apply. Comparing histories $(0, 0)$ and $(0, 1)$ gives

$$G(0) = z[L_R - L^{C(0)}] + (p + c_2 - z)[W_R - W^{C(0)}].$$

The war and operation payoffs are identical at these histories, while settlement rises from $-a$ to zero. Under the common continuation tie rule, every type either takes the same action or

replaces force with settlement. Conceding zero therefore weakly improves the target's payoff under every posterior.

To prove the strict comparison when $z < \lambda\bar{c}$, let $t = \max\{p, \frac{z}{\lambda}\}$. Assumption 1(ii) gives $p < \bar{c}$, so $t < \bar{c}$. Every type in

$$\left(t, \min\left\{\bar{c}, \max\left\{p + a, \frac{z + a}{\lambda}\right\}\right\}\right)$$

settles after compliance with zero but uses force after resistance. This interval has positive length because $a > 0$ raises both thresholds strictly. Since $p + c_2 > 0$ and $z > 0$, these types strictly improve the target's payoff and all others weakly improve it, so $G(0) > 0$. Continuity then makes a positive demand acceptable under the prior.

Conversely, if $z \geq \lambda\bar{c}$, no supported type settles after compliance with zero under the tie rule, and none settles after resistance. The same war-operation comparison applies after both responses, so $G(0) = 0$. The strict decrease of G on the uncapped domain and rejection of capped demands imply $x^* = \Omega^* = 0$. This proves Remark 3 without imposing the closed-form conditions.

In Proposition 2's region, $z < \lambda\bar{c}$ and the largest accepted demand is positive. Strict monotonicity and rejection of capped demands give a unique largest accepted demand x^* . Where the operation is active after both responses to zero and $\kappa_{L,R} \geq \bar{c}$, the war cutoffs are equal, $L_R = 1$, and $L^{C(0)} = F(\frac{z}{\lambda})$, so the expression for $G(0)$ reduces to $G(0) = z\alpha$. This simplification is specific to that branch; the positive-demand argument above also covers a dominated operation after compliance with zero.

Equilibrium strategies. Lemma 2 establishes that no coercer type prefers resistance. In the operation game, every type demands x^* , the target concedes, and types below $\kappa_W(x^*)$ fight, so war occurs with probability $W(x^*)$. In the benchmark, every type demands $x^\dagger$, and types below $\kappa^\dagger(x^\dagger)$ fight.

Intuitive criterion. In the benchmark, the optimized continuation payoffs are

$$U_C^{\dagger(x, c_1)} = \max\{x, p - c_1\}, \quad U_R^{\dagger(c_1)} = \max\{-a, p - c_1\}.$$

In the operation game they are

$$U_{C(x,c_1)} = \max\{x, p - c_1, \min\{x + z, 1\} - \lambda c_1\}, \quad U_{R(c_1)} = \max\{-a, p - c_1, z - \lambda c_1\}.$$

In either game U_C is weakly increasing over the entire demand set, including capped demands. Each continuation action pays weakly more after concession than after resistance, including at zero, so $U_{R(c_1)} \leq U_{C(x^e, c_1)}$. The preceding acceptance argument identifies x^e as the largest demand accepted under the prior. The construction in SI §C.1. [Off-Path Beliefs and the Intuitive Criterion](#) therefore completes both equilibria. It also proves that the selected zero-demand outcome when $z \geq \lambda \bar{c}$ satisfies the criterion, covering Remark 3 without interior cutoffs. All comparisons evaluated at these selected equilibria inherit this same belief construction.

Stronger refinement. §E.3. [D1 and the Selected Pooling Equilibria](#) gives the D1 calculation for these outcomes.

Closed form in the operation game. In the active, interior region, $W = \frac{p-x-z}{(1-\lambda)\bar{c}}$, $W_0 = \frac{p-z}{(1-\lambda)\bar{c}}$, and $W_0 - W = \frac{x}{(1-\lambda)\bar{c}}$. Moreover, $L = \frac{z}{\lambda\bar{c}} = 1 - \alpha$ is constant in x . If $\kappa_{L,R} \geq \bar{c}$, every nonfighting type uses the operation after resistance. Hence $L_0 = 1$ and $L_0 - L = \alpha$. The binding acceptance condition becomes

$$\frac{x((1-\lambda)\bar{c} - p + x + z)}{(1-\lambda)\bar{c}} = \frac{x(p + c_2 - z)}{(1-\lambda)\bar{c}} + z\alpha$$

Rearranging yields $x^2 + x((1-\lambda)\bar{c} - 2p - c_2 + 2z) - (1-\lambda)\bar{c}z\alpha = 0$. Its constant term is negative because $\alpha > 0$. The unique positive root is therefore (9).

$$x^* = \frac{1}{2} \left[-((1-\lambda)\bar{c} - 2p - c_2 + 2z) + \sqrt{((1-\lambda)\bar{c} - 2p - c_2 + 2z)^2 + 4(1-\lambda)\bar{c}z\alpha} \right] \quad (9)$$

□

D. PROOFS OF THE COMPARATIVE STATICS

§A.1. [Results, Conditions, and Proofs](#) locates these derivations by the results they establish. §D.10. [Interior Switch Accounting](#) supplies the switching arithmetic and joint-loss conditions used in the threat–assurance comparison.

D.1. Proof of Remark 1

Proof. At the largest accepted demand, (7) binds, so $V_2(x^*) = V_{2,R}$. Let $W = W(x^*)$, $L = L(x^*)$, $W_0 = W_R$, and $L_0 = L_R$. Substituting the target's payoffs yields

$$W(1 - p - c_2) + (1 - x^*)(1 - W) - S(x^*)(L - W) = W_0(1 - p - c_2) + (1 - W_0) - z(L_0 - W_0)$$

Cancel common terms and collect the terms in x^* :

$$x^*(1 - W) = (p + c_2)(W_0 - W) + z(L_0 - W_0) - S(x^*)(L - W)$$

Separating $(p + c_2)W_0$ from $(p + c_2)W$ gives the expanded identity

$$\begin{aligned} \Omega^* = & \underbrace{(p + c_2)W_R + z(L_R - W_R)}_{\text{threat: punishment after resistance} =: \Theta^*} \\ & - \underbrace{[(p + c_2)W(x^*) + S(x^*)(L(x^*) - W(x^*))]}_{\text{assurance failure: harm not prevented by compliance} =: \Delta^*} \end{aligned} \quad (10)$$

Definition 1 gives $\Omega^* = x^*(1 - W)$, so this is (3). Coercive ability is zero if and only if $\Theta^* = \Delta^*$. Subtracting the benchmark identity gives

$$\Omega^* - \Omega^\dagger = (\Theta^* - \Theta^\dagger) - (\Delta^* - \Delta^\dagger) \quad (11)$$

Response to punishment. In the closed-form region, hold the post-compliance continuation rules fixed as functions of the demand. Differentiating the binding acceptance condition with respect to Θ^* gives

$$\frac{\partial x^*}{\partial \Theta^*} = \frac{1}{1 - W(x^*) - \frac{p + c_2 - z - x^*}{(1 - \lambda)\bar{c}}}$$

The denominator is positive because the acceptance margin decreases in the demand. It is less than one when $x^* < p + c_2 - z$, so the derivative then exceeds one. This partial derivative isolates a change in punishment; changing an operation parameter can also change the post-compliance continuation schedule. $\square$

D.2. Proof of Proposition 3

Proof. Evaluate a common conceded demand under the prior, and suppose the operation is active with an interior operation–settlement cutoff. By Lemma 3, the coercer settles when $c_1 > \kappa_L(x^*)$. Thus $\alpha = 1 - F(\kappa_L(x^*))$.

At every accepted demand, Assumption 1(i) makes $S(y) = z$. Settlement pays y , while the operation pays $y + z - \lambda c_1$. The actions are equally valuable at $\kappa_L(y) = \frac{z}{\lambda}$, which is constant in the transfer. Consequently, $\alpha = 1 - F(\frac{z}{\lambda})$. Since F is strictly increasing at this cutoff, $\frac{\partial \alpha}{\partial z} = -\frac{f(\frac{z}{\lambda})}{\lambda} < 0$ and $\frac{\partial \alpha}{\partial \lambda} = f(\frac{z}{\lambda}) \frac{z}{\lambda^2} > 0$. $\square$

D.3. Proof of Corollary 1

Proof. At history (y, r) , settlement pays $y - a\mathbb{1}[r = 1]$ and the operation pays $y + g - \lambda c_1$. Their difference is $g + a\mathbb{1}[r = 1] - \lambda c_1$, which vanishes at $\kappa_L(y, r) = \frac{g + a\mathbb{1}[r = 1]}{\lambda}$. After compliance, settlement also has to beat war, so the coercer settles exactly when $c_1 > \max\{p - x, \frac{g}{\lambda}\}$. This gives the stated probability. If the operation is active, $\frac{g}{\lambda} > p - x$, so the expression reduces to $1 - F(\frac{g}{\lambda})$. Target harm η does not enter either continuation comparison.

Now set $g = 0$. The operation pays $x - \lambda c_1$ against x from settlement after compliance, so no positive-cost type conducts it. War remains available: $W(x) = L(x) = F(p - x)$ and $\alpha = 1 - F(p - x)$. After resistance, the operation pays $-\lambda c_1$ against $-a$ from settlement and $p - c_1$ from war. Its cutoffs are $\kappa_{W,R} = \frac{p}{1-\lambda}$ and $\kappa_{L,R} = \frac{a}{\lambda}$, and it is active if and only if $a > \lambda(p + c_1)$ under the stated interior conditions. The games share the post-compliance node, so the target's loss from conceding x is the same strictly increasing function in both. Their largest accepted demands therefore have the same ordering as their expected resistance losses. Those losses are $(p + c_2)\frac{p+a}{\bar{e}}$ in the benchmark and $(p + c_2)\frac{p}{(1-\lambda)\bar{e}} + \frac{\eta(\frac{a-p}{1-\lambda})}{\bar{e}}$ with the operation. Their difference has the sign of $(a(1-\lambda) - \lambda p)(\eta - \lambda(p + c_2))$, whose first factor is positive by operation activity. Coercive ability $x[1 - F(p - x)]$ also strictly increases in x . Hence both the accepted demand and coercive ability rise exactly when $\eta > \lambda(p + c_2)$.

To complete the selected equilibria under the intuitive criterion, note that $U_{C(x, c_1)} = \max\{x, p - c_1, x + g - \lambda c_1\}$ is weakly increasing throughout the stated feasible demand set. Resistance gives $U_{R(c_1)} = \max\{-a, p - c_1, g - \lambda c_1\} \leq U_{C(x, c_1)}$ for $x \geq 0$. Harm η changes the

target's acceptance condition but neither of these incentive properties. Apply SI §C.1. Off-Path Beliefs and the Intuitive Criterion to each game's largest prior-accepted demand. Thus the demand and coercive-ability comparison at $g = 0$ holds under the maintained refinement. $\square$

D.4. Proof of Proposition 6

With interior cutoffs and an active operation after concession, the expected payoffs are

$$\begin{aligned} V^\dagger &= x^\dagger + \frac{(p - x^\dagger)^2}{2\bar{c}} \\ V^* &= x^* + \frac{(p - x^* - z)^2}{2(1 - \lambda)\bar{c}} + \frac{z^2}{2\lambda\bar{c}} \end{aligned} \tag{12}$$

Proof. Benchmark. Types below $\kappa^\dagger = p - x^\dagger$ fight and receive $p - c_1$. The remaining types settle and receive $x^\dagger$. Under the uniform distribution,

$$V^\dagger = \frac{1}{\bar{c}} \left(\int_0^{\kappa^\dagger} (p - c) dc + \int_{\kappa^\dagger}^{\bar{c}} x^\dagger dc \right) = x^\dagger + \frac{1}{\bar{c}} \kappa^\dagger \left(p - \frac{\kappa^\dagger}{2} - x^\dagger \right)$$

Since $p - x^\dagger = \kappa^\dagger$, the term in parentheses is $\frac{\kappa^\dagger}{2}$. Substitution gives the first line of (12).

Operation game. Types below κ_W fight. Types in $(\kappa_W, \kappa_L]$ use the operation and receive $x^* + z - \lambda c_1$. The remaining types settle. Integrating over the three regions gives

$$V^* = x^* + \frac{1}{\bar{c}} \left(\kappa_W(p - x^* - z) - (1 - \lambda) \frac{\kappa_W^2}{2} + z\kappa_L - \lambda \frac{\kappa_L^2}{2} \right)$$

Since $p - x^* - z = (1 - \lambda)\kappa_W$, the first two terms reduce to $\frac{(1 - \lambda)\kappa_W^2}{2}$. Since $\kappa_L = \frac{z}{\lambda}$, the remaining operation term becomes $\frac{z^2}{2\lambda}$. Substituting $\kappa_W = \frac{p - x^* - z}{1 - \lambda}$ yields the second line of (12).

Let $Q_0(y) = y + \frac{(p - y)^2}{2\bar{c}}$ denote the benchmark continuation value at transfer y , and let $Q_1(y) = y + \frac{(p - y - z)^2}{2(1 - \lambda)\bar{c}} + \frac{z^2}{2\lambda\bar{c}}$ denote the operation-game value on its active interior branch. An interior operation-game war cutoff at x^* implies $p - x^* > z > 0$, while Assumption 1(ii) implies $p - x^* < \bar{c}$. The benchmark cutoff evaluated at x^* is therefore interior. Adding and subtracting $Q_0(x^*)$ and simplifying gives

$$V^* - V^\dagger = \underbrace{\frac{[\lambda(p - x^*) - z]^2}{2\lambda(1 - \lambda)\bar{c}}}_{\text{menu at the same transfer}} + \underbrace{Q_0(x^*) - Q_0(x^\dagger)}_{\text{endogenous-demand response}}$$

In particular,

$$Q_1(x^*) - Q_0(x^*) = \frac{(\lambda(p - x^*) - z)^2}{2\lambda(1 - \lambda)\bar{c}} \geq 0$$

which establishes the nonnegative menu gain. Moreover, $Q_{0'}(y) = 1 - \frac{p-y}{\bar{c}} > 0$ whenever the benchmark cutoff is interior. Thus $Q_0(x^*) - Q_0(x^\dagger) < 0$ if $x^* < x^\dagger$. The equilibrium payoff falls exactly when this loss exceeds the menu gain. $\square$

D.5. Proof of Proposition 7

Proof. On the selected equilibrium path, the target concedes. Total conflict cost is zero under settlement, $c_1 + c_2$ under war, and λc_1 under the operation. The demand x and the operation's gain z do not enter because each is transferred between the states.

Let $b = \kappa^\dagger(x^\dagger) = p - x^\dagger$. If the operation is evaluated after a concession $x^\dagger - d$, its war cutoff is

$$\kappa_W(x^\dagger - d) = \frac{p - x^\dagger + d - z}{1 - \lambda} = \frac{b + d - z}{1 - \lambda}$$

while $\kappa_L(x^\dagger - d) = \frac{z}{\lambda}$ is constant. At $d = d_W$, the war cutoff equals b and the operation remains active because $b < \frac{z}{\lambda}$. Below this threshold, $\kappa_W(x^\dagger - d) < b$ and war falls. Above it, the active-branch war cutoff exceeds b ; if the operation later becomes dominated, the war cutoff becomes $p - (x^\dagger - d) = b + d > b$. Hence the subwar capability lowers war exactly when $d < z - \lambda b = d_W$.

While the operation is active, conflict cost after concession $x^\dagger - d$ is

$$K^L(d) = \int_0^{\kappa_W(x^\dagger - d)} (c_1 + c_2) dF(c_1) + \lambda \int_{\kappa_W(x^\dagger - d)}^{\kappa_L(x^\dagger - d)} c_1 dF(c_1)$$

and $\Delta K(d) = K^{L(d)} - K^\dagger$. Differentiating with respect to d gives

$$\frac{d\Delta K(d)}{dd} = f(\kappa_W(x^\dagger - d)) \left[\kappa_W(x^\dagger - d) + \frac{c_2}{1 - \lambda} \right] > 0 \quad (13)$$

because $\frac{d\kappa_W(x^\dagger - d)}{dd} = \frac{1}{1-\lambda}$ and the upper operation cutoff is constant.

If the operation becomes dominated, the continuation menu reduces to war and settlement with cutoff $b + d$. Conflict cost is then $\int_0^{b+d} (c_1 + c_2) dF(c_1)$, whose derivative in d is $f(b + d)(b + d + c_2) > 0$. Conflict cost is continuous where the operation leaves the menu because the interval of types using it shrinks to zero measure. Thus $\Delta K(d)$ is strictly increasing throughout the stated comparison range. The interior restriction matters: before any type fights, a smaller concession can leave conflict cost unchanged.

At $d = d_W$, the two war cutoffs coincide: $\kappa_W(x^\dagger - d_W) = b$. Types below b fight in both games, while types between b and $\kappa_L(x^\dagger - d_W)$ replace benchmark settlement with the operation. Hence

$$\Delta K(d_W) = \lambda \int_b^{\kappa_L(x^\dagger - d_W)} c_1 dF(c_1) > 0 \quad (14)$$

Since $\Delta K(d)$ is continuous and strictly increasing, $\Delta K(0) < 0$ implies a unique root $d_K \in (0, d_W)$. Below this root the direct cost saving remains; above it the concession loss raises total conflict cost beyond the benchmark. If $\Delta K(0) \geq 0$, strict monotonicity gives $\Delta K(d) > 0$ for every $d > 0$.

Finally, $b > 0$ and $\lambda > 0$ imply $d_W = z - \lambda b < z$. Thus $d \geq z$ implies $d > d_W$ and therefore $\Delta K(d) > 0$.

Local relative-cost comparison. Now vary the operation's relative cost without imposing the preceding path conditions. Let $q = 1 - \lambda$ and $v(x) = p - x - z$. On an active interior branch with a uniform cost distribution,

$$K(x, \lambda) = \frac{1}{2\bar{c}} \left[v(x) \frac{v(x) + 2c_2}{q} + \frac{z^2}{\lambda} \right].$$

Holding the concession fixed and differentiating gives

$$\frac{\partial K(x, \lambda)}{\partial \lambda} = \frac{1}{2\bar{c}} \left[v(x) \frac{v(x) + 2c_2}{q^2} - \frac{z^2}{\lambda^2} \right].$$

This derivative is negative if and only if

$$\left(\frac{z}{\lambda}\right)^2 > \frac{v(x)(v(x) + 2c_2)}{(1 - \lambda)^2}. \quad (15)$$

Thus (15) is necessary and sufficient for lower operating cost to raise conflict cost at a fixed concession. When $c_2 = 0$, operation activity requires $\frac{v(x)}{q} < \frac{z}{\lambda}$, which is exactly the strict inequality needed to make the derivative negative.

The equilibrium demand response. Maintain the active, interior closed-form regime and the strict restriction $z + a > \lambda\bar{c}$. The positive root in (9) is differentiable on this open branch. Its defining quadratic, with $\alpha = 1 - \frac{z}{\lambda\bar{c}}$, gives

$$\frac{\partial x^*}{\partial \lambda} = \frac{\bar{c}(x^* - z) + \frac{z^2}{\lambda^2}}{2x^* + (1 - \lambda)\bar{c} - 2p - c_2 + 2z}. \quad (16)$$

The denominator is positive at the positive root. The interior post-compliance war cutoff implies $p + c_2 > z$ and $0 < W(x^*) < 1$. Proposition 4 and $x^* > 0$ therefore give

$$x^* > \Omega^* = \frac{x^*(p + c_2 - z)}{(1 - \lambda)\bar{c}} + z\alpha > z\alpha.$$

Thus the numerator in (16) satisfies

$$\bar{c}(x^* - z) + \frac{z^2}{\lambda^2} > \bar{c}(z\alpha - z) + \frac{z^2}{\lambda^2} = (1 - \lambda)\frac{z^2}{\lambda^2} > 0.$$

Hence $\frac{\partial x^*}{\partial \lambda} > 0$: cheaper operation lowers the accepted concession on this branch.

Conflict cost also satisfies

$$\frac{\partial K(x, \lambda)}{\partial x} = -\frac{v(x) + c_2}{q\bar{c}} < 0.$$

Along the equilibrium branch,

$$\frac{\partial K^*}{\partial \lambda} = \frac{\partial K(x^*, \lambda)}{\partial \lambda} + \frac{\partial K(x^*, \lambda)}{\partial x} \frac{\partial x^*}{\partial \lambda}.$$

If (15) holds at $x = x^*$, the first term is strictly negative and the second is strictly negative by the derived demand response. Therefore $\frac{\partial K^*}{\partial \lambda} < 0$. When $c_2 = 0$, activity itself implies (15), proving the unconditional sign within this region.

At $z + a = \lambda \bar{c}$, decreasing λ preserves the branch on which no type settles after resistance. The same expressions and strict signs therefore apply to the left derivatives. A change through the boundary in the other direction changes the resistance-stage formula, so no derivative sign on that side follows from this proof. $\square$

PROPOSITION 8 (The Post-Compliance Cutoff Is Selection-Free): Consider any equilibrium and any demand conceded with positive probability. If the operation is active after that demand and $S(x) = z$, the coercer settles if and only if $c_1 > \frac{z}{\lambda}$. This cutoff is the same at every such demand and under every off-path belief. If the operation is active and $S(x) = z$ after every conceded demand, so those demands exceed $p - \frac{z}{\lambda}$, then conditional on compliance the settling types are $(\frac{z}{\lambda}, \bar{c}]$ in every equilibrium.

D.6. Proof of Proposition 8

Proof. Fix an equilibrium and a conceded demand $x \geq 0$, at history $(y, r) = (x, 0)$. The coercer then chooses among settlement, war, and the operation, with payoffs x , $p - c_1$, and $x + S(x) - \lambda c_1$. These payoffs depend only on c_1 , x , and the model parameters. Because the target has already moved, they do not depend on its beliefs or the demands made by other types.

The operation payoff minus the settlement payoff is $S(x) - \lambda c_1$. When $S(x) = z$, this difference is $z - \lambda c_1$ and does not depend on x . The operation is preferred to settlement exactly when $c_1 < \frac{z}{\lambda}$. Because the operation is active by hypothesis, Lemma 3 implies that types above $\kappa_L(x) = \frac{z}{\lambda}$ settle. If μ_x is the posterior at this demand, its conditional settlement probability is $1 - \mu_x([0, \frac{z}{\lambda}])$. This equals $1 - F(\frac{z}{\lambda})$ in a pooling equilibrium.

The cutoff is therefore $\frac{z}{\lambda}$ at every conceded demand for which the operation is active. Activity itself depends on the demand. Equation (6) implies that the operation enters the coercer's envelope after concession of x exactly when $x > p - \frac{z}{\lambda}$. In a separating equilibrium with two accepted demands, the operation may be active after the larger demand and dominated after the smaller one. The corresponding settlement cutoffs would then be $\frac{z}{\lambda}$ and $\kappa_S(x_L)$. If every conceded demand satisfies $S(x) = z$ and exceeds $p - \frac{z}{\lambda}$, the pointwise cutoff is common

across those histories. Its conditional probability still depends on the posterior, and types facing resistance need not settle. $\square$

D.7. Proof of The Resistance-Stage War Comparison

Proof. If the operation is active after resistance, the coercer fights when $c_1 \leq \kappa_{W,R}$. If it is dominated, the coercer fights when $c_1 \leq \kappa_{S,R} = p + a$. After resistance, $S(0) = z$ and the audience cost applies. Thus $\kappa_{W,R} = \frac{p-z}{1-\lambda}$ and $\kappa_{L,R} = \frac{z+a}{\lambda}$, which gives the two probabilities stated in the main text. The operation is active if and only if $\kappa_{W,R} < \kappa_{L,R}$, or

$$\lambda(p - z) < (1 - \lambda)(z + a)$$

Rearranging gives $\lambda(p + a) < z + a$. The condition for the operation to lower war is $\kappa_{W,R} < \kappa_{S,R} = p + a$, which rearranges to the same inequality. Thus the operation is active exactly when it lowers war. [Assumption 1\(ii\)](#) implies $\kappa_{W,R} \leq \frac{p}{1-\lambda} < \bar{c}$, so the reduction is strict whenever the operation enters the menu. The operation never raises war after resistance. The extension in which the operation carries its own audience cost is derived in [§F.3. The Operation's Audience Cost](#). $\square$

D.8. Proof of Proposition 4

Proof. Joint deterioration. Under the stated closed-form conditions, [§D.10. Interior Switch Accounting](#) derives (20) as equivalent to weaker punishment and greater post-compliance harm. Substitution into (11) gives strictly lower coercive ability. The following calculations give the comparison when the two components need not both deteriorate.

Benchmark. With an interior war cutoff, $W^\dagger(x^\dagger) = \frac{p-x^\dagger}{\bar{c}}$. Therefore

$$\Omega^\dagger = x^\dagger(1 - W^\dagger) = \frac{x^\dagger(\bar{c} - p + x^\dagger)}{\bar{c}}$$

Equation (8) implies $x^{\dagger 2} = (p + c_2)a - x^\dagger(\bar{c} - 2p - c_2)$. Substitute this expression into the numerator:

$$x^\dagger(\bar{c} - p) + x^{\dagger 2} = x^\dagger(\bar{c} - p) + (p + c_2)a - x^\dagger(\bar{c} - 2p - c_2) = (p + c_2)(x^\dagger + a)$$

Dividing by $\bar{c}$ gives $\Omega^\dagger = (p + c_2)\frac{x^\dagger + a}{\bar{c}}$.

The benchmark quadratic also gives

$$\frac{\partial x^\dagger}{\partial p} = \frac{2x^\dagger + a}{\bar{c} - 2p - c_2 + 2x^\dagger} > 0, \quad \frac{\partial(p - x^\dagger)}{\partial p} = \frac{\bar{c} - 2p - c_2 - a}{\bar{c} - 2p - c_2 + 2x^\dagger}.$$

The denominator is positive by Assumption 1(ii). The derivative of the war cutoff need not be positive, but

$$\frac{\partial \Omega^\dagger}{\partial p} = \frac{x^\dagger + a + (p + c_2) \frac{\partial x^\dagger}{\partial p}}{\bar{c}} > 0.$$

Thus equilibrium coercive ability rises with power even when the equilibrium war probability does not.

Operation game. In the active, interior region, $W(x^*) = \frac{p - x^* - z}{(1 - \lambda)\bar{c}}$. Hence $\Omega^* = \frac{x^*((1 - \lambda)\bar{c} - p + x^* + z)}{(1 - \lambda)\bar{c}}$. Equation (9) gives $x^{*2} = (1 - \lambda)\bar{c}z\alpha - x^*((1 - \lambda)\bar{c} - 2p - c_2 + 2z)$. Substituting yields

$$\begin{aligned} & x^*((1 - \lambda)\bar{c} - p + z) + x^{*2} \\ &= x^*((1 - \lambda)\bar{c} - p + z) + (1 - \lambda)\bar{c}z\alpha \\ & \quad - x^*((1 - \lambda)\bar{c} - 2p - c_2 + 2z) \\ &= x^*(p + c_2 - z) + (1 - \lambda)\bar{c}z\alpha \end{aligned}$$

Divide by $(1 - \lambda)\bar{c}$ to obtain $\Omega^* = \frac{x^*(p + c_2 - z)}{(1 - \lambda)\bar{c}} + z\alpha$. Since $W_R - W(x^*) = \frac{x^*}{(1 - \lambda)\bar{c}}$, this also gives (4). Comparing with the benchmark, coercive ability is lower exactly when

$$(p + c_2) \frac{x^\dagger + a}{\bar{c}} > \frac{x^*(p + c_2 - z)}{(1 - \lambda)\bar{c}} + z\alpha \quad (17)$$

Both demands are given in primitives by (8) and (9). The protection term $z\alpha = z(1 - \frac{z}{\lambda\bar{c}})$ is single-peaked in reach, with its maximum at $z = \lambda\frac{\bar{c}}{2}$. Its derivative is $1 - 2\frac{z}{\lambda\bar{c}}$ on this branch. $\square$

D.9. Proof of Proposition 5

Proof. Every-type payoff comparison. At the two common accepted demands, the continuation payoffs are

$$U^\dagger(c_1) = \max\{p - c_1, x^\dagger\}, \quad U^*(c_1) = \max\{p - c_1, x^*, x^* + z - \lambda c_1\}.$$

Since $b = p - x^\dagger < \bar{c}$, a positive measure of benchmark types settles. If $d < 0$, each of these types can receive $x^* > x^\dagger$ in the operation game, so every-type weak loss requires $d \geq 0$. Given $d \geq 0$, the benchmark weakly dominates the operation game's war and settlement payoffs. The operation's advantage over the benchmark payoff increases with c_1 below b , with slope $1 - \lambda$, and decreases above b , with slope $-\lambda$. Its maximum is attained at b and equals

$$x^* + z - \lambda b - x^\dagger = z - \lambda b - d.$$

Hence $U^*(c_1) \leq U^\dagger(c_1)$ for every type exactly when $d \geq 0$ and $d \geq z - \lambda b$, which is (5).

War after compliance. War must beat both settlement and the operation. Its actual cutoff in the operation game is therefore

$$k = \min \left\{ b + d, \frac{b + d - z}{1 - \lambda} \right\}, \quad W(x^*) = F(k).$$

This formula includes a dominated operation and cutoffs outside the support through the clamped CDF. Because b is interior, $F(k) \geq F(b)$ if and only if $k \geq b$, and $F(k) > F(b)$ if and only if $k > b$. The weak comparison requires both $d \geq 0$ and $d \geq z - \lambda b$; the strict comparison requires both inequalities to be strict. This proves the weak and strict war statements.

Strict consequences of a positive lost concession. Suppose (5) holds and $d > 0$. For every $c_1 > b$, settlement in the operation game pays less than $x^\dagger$, war pays $p - c_1 < x^\dagger$, and the operation pays

$$x^* + z - \lambda c_1 \leq x^\dagger - \lambda(c_1 - b) < x^\dagger.$$

Thus every benchmark settler strictly loses. Types at or below b fight in both games and incur the same conflict costs. If $k > b$, a positive measure of benchmark settlers now fights, adding strictly positive costs; all other benchmark settlers incur nonnegative costs. If $k = b$, then $d = z - \lambda b > 0$ and $\frac{z}{\lambda} > b$. Types in $(b, \min\{\frac{z}{\lambda}, \bar{c}\})$ operate instead of settling, again adding strictly positive conflict costs. Thus expected joint conflict cost rises in either case. Finally,

$$\Omega^* = x^*[1 - F(k)] \leq x^*[1 - F(b)] < x^\dagger[1 - F(b)] = \Omega^\dagger,$$

where the strict inequality uses $x^* < x^\dagger$ and $F(b) < 1$. If $d = 0$ and (5) holds, the operation never strictly improves on the benchmark continuation menu. Under the maintained tie rule the benchmark actions, payoffs, and all stated outcomes are unchanged.

A stronger sufficient test in the closed-form region. The condition $d > z$ is sufficient for the strict frontier because $\max\{0, z - \lambda b\} < z$. Under the additional closed-form conditions of Propositions 1 and 2, it can be checked directly in primitives:

$$(p + c_2)a + z(2p + c_2 - (1 - \lambda)\bar{c} - z - (1 - \lambda)\bar{c}\alpha) > \lambda\bar{c}x^\dagger. \quad (18)$$

To establish the equivalence between this test and $d > z$, define

$$\varphi(t) = t^2 + [(1 - \lambda)\bar{c} - 2p - c_2 + 2z]t - (1 - \lambda)\bar{c}z\alpha.$$

Its unique positive root is x^* . Assumption 1(ii) implies $\varphi(-z) = z[2p + c_2 - (1 - \lambda)\bar{c} - z - (1 - \lambda)\bar{c}\alpha] < 0$. Since $x^\dagger - z \geq -z$, the argument $x^\dagger - z$ cannot lie below the negative root. Therefore $d > z$ if and only if $\varphi(x^\dagger - z) > 0$. Substituting the benchmark quadratic gives

$$\varphi(x^\dagger - z) = (p + c_2)a + z[2p + c_2 - (1 - \lambda)\bar{c} - z - (1 - \lambda)\bar{c}\alpha] - \lambda\bar{c}x^\dagger,$$

which proves (18). This polynomial test remains valid when its argument is negative or outside the active operation branch; it does not evaluate an off-branch continuation probability. $\square$

D.10. Interior Switch Accounting

Hold transfer y fixed, suppose $S(y) = z$, and let every continuation cutoff be interior. Write $A = a\mathbb{1}[r = 1]$ at that history and define the operation's entry margin

$$M(y, r) = z + (1 - \lambda)a\mathbb{1}[r = 1] - \lambda(p - y).$$

Write $M(y)$ for the margin at the fixed history and $M_R = M(0, 1)$. The operation enters the coercer's menu when $M(y) > 0$. Relative to the benchmark, the masses that switch from war and settlement to the operation are

$$m_{W(y)} = \frac{M(y)}{(1 - \lambda)\bar{c}}, \quad m_{S(y)} = \frac{M(y)}{\lambda\bar{c}}.$$

The first equality follows from $\kappa_S(y) - \kappa_W(y) = \frac{M(y)}{1 - \lambda}$ and the second from $\kappa_L(y) - \kappa_S(y) = \frac{M(y)}{\lambda}$.

A former fighter lowers continuation harm excluding the retained transfer by $p + c_2 - z$, while a former settler raises that component by z . Its expected change at history y is therefore

$$zm_S(y) - (p + c_2 - z)m_W(y) = \frac{M(y)(z - \lambda(p + c_2))}{\lambda(1 - \lambda)\bar{c}}. \quad (19)$$

This expression has the sign of $z - \lambda(p + c_2)$. The change in total target loss at the same transfer also includes $ym_{W(y)}$, because replacing war preserves the transfer; at resistance $y = 0$, so the two comparisons coincide. Applying the continuation-harm comparison after both target responses and using $M_R - M(x) = (1 - \lambda)a - \lambda x$ shows that the change in threat relative to the change in assurance has the sign of $(z - \lambda(p + c_2))((1 - \lambda)a - \lambda x)$. The first factor determines whether the operation raises harm at a given history, and the second determines whether more types switch after resistance or compliance.

The closed-form region truncates the settlement-to-operation switch after resistance because $\kappa_{L,R} \geq \bar{c}$. There, $\Theta^* < \Theta^\dagger$ and $\Delta^* > \Delta^\dagger$ hold together exactly when

$$(1 - \lambda)(\bar{c} - p - a)z < M_{R(p+c_2-z)}, \quad \frac{M(x^*)(z - \lambda(p + c_2))}{\lambda(1 - \lambda)} + (p + c_2)(x^\dagger - x^*) > 0. \quad (20)$$

The first inequality says that war replacement reduces punishment after resistance despite the operation's effect on former settlers. The second says that post-compliance harm rises once the smaller accepted demand is included. This is an exact conditional statement, not a claim about prevalence in a parameter population.

E. PURE TARGET RESPONSES AND D1

Pure target responses permit an interval of equilibrium demands, including zero under the conditions below. This section characterizes that interval under the intuitive criterion before considering D1.

Write $\ell = \frac{z}{\lambda}$ and $s_R = \max\{p + a, \frac{z+a}{\lambda}\}$. In each game, let $u_2^C(d, c_1)$ and $u_{2,R}(c_1)$ be the target's payoffs under the coercer's optimal continuation choices. Define the unnormalized prior acceptance surplus and its largest accepted demand by

$$\Phi(d) = \int_0^{\bar{c}} (u_2^C(d, c_1) - u_{2,R}(c_1)) dc_1 \quad x^e = \max\{d \in [0, 1] : \Phi(d) \geq 0\}$$

The lower endpoint is

$$\underline{x} = \begin{cases} 0 & \text{benchmark: } \bar{c} > p + a \\ p + c_2 & \text{benchmark: } \bar{c} \leq p + a \\ 0 & \text{operation: } \bar{c} > s_R \text{ or } z > \lambda p \\ \min\{z, p + c_2 - z\} & \text{operation: } \bar{c} \leq s_R \text{ and } z \leq \lambda p \end{cases}$$

Under the conditions of Propositions 1 and 2, $x^e = x^\dagger$ and $x^e = x^*$, respectively.

LEMMA 5 (Equilibrium Outcomes Under Pure Target Responses): Under [Assumption 1](#) and the maintained tie rules, the possible largest accepted demands among perfect Bayesian equilibria with pure target responses are exactly $x \in [\underline{x}, x^e]$. The same interval obtains under the intuitive criterion. Each value admits a pooling equilibrium, and every equilibrium is equivalent to its corresponding pool in each type's military action and both players' payoffs. Only types satisfying $U_C(x, c_1) = U_R(c_1)$ can use another signal. Allowing the coercer to randomize over demands adds no terminal outcomes.

Proof. I establish the two bounds and construct supporting beliefs. First partition types by their resistance action and their preferred nonwar action after uncapped compliance. Write W, O, S for war, operation, and settlement. The pairs and associated target thresholds are

$$(W, O) : p + c_2 - z, \quad (O, O) : 0, \quad (W, S) : p + c_2 \quad (O, S) : z, \quad (S, S) : 0$$

Retain nonempty regions, including singleton types under the continuation tie rule; the benchmark has only (W, S) and (S, S). If compliance induces war, the target's surplus is zero. Otherwise it is $J - d$, where J is the corresponding threshold. Hence every posterior requires concession when $d \leq \min J$. Conversely, in a region attaining the minimum, nonwar compliance becomes optimal by $d = J$ at the latest: its threshold against resistance war is at most $p - z \leq p + c_2 - z$ for the operation and $p \leq p + c_2$ for settlement. After resistance operation or settlement, a nonwar compliance action is already optimal at zero. Every $d > \min J$ therefore

admits a strictly rejecting posterior. Capped demands are strictly rejected under every posterior by Assumption 1(i).

In the operation game, resistance settlement exists exactly when $\bar{c} > s_R$, and operation after resistance and compliance with zero exists exactly when $z > \lambda p$. Assumption 1(ii) puts $b_R = \frac{p-z}{1-\lambda}$ below $\bar{c}$. Without either region, $\ell \leq p \leq b_R < \bar{c}$, giving $\min J = \min\{z, p + c_2 - z\}$. In the benchmark, settlement after resistance exists exactly when $\bar{c} > p + a$. These observations give $\min J = \underline{x}$, including the strict endpoint inequalities.

The prior surplus strictly decreases on the uncapped demand domain. Where war occurs, the derivative of unnormalized compliance loss is $\bar{c} - 2p - c_2 + 2d$ on the benchmark or dominated-operation branch and $\bar{c} - \frac{2p+c_2-2z-2d}{1-\lambda}$ on the active-operation branch. Assumption 1(ii) makes both positive. After war disappears the derivative is $\bar{c}$, and loss is continuous across branches. Since $\Phi(0) \geq 0$ and surplus is negative at demand 1 in the benchmark and $1 - z$ in the operation game, x^e is its unique nonnegative zero. The prior accepts exactly $[0, x^e]$; the universal lower bound implies $\Phi(\underline{x}) \geq 0$.

Every equilibrium has a largest accepted demand. Zero must be accepted. Without a maximum, the positive mass of high-cost nonwar types would face an unattained payoff supremum along accepted demands. Randomization cannot attain that supremum either. At the maximum x , each type receives $U_C(x, c_1)$: smaller accepted demands and resistance pay no more. For $x > 0$, every type with a strict compliance gain uniquely demands x . All other types fight after either response, contributing zero to the acceptance-surplus numerator. Removing or reweighting them leaves its sign unchanged, so Bayes' rule requires $\Phi(x) \geq 0$. At $x = 0$ this inequality already holds. Thus $\underline{x} \leq x \leq x^e$.

For sufficiency, pool at any such x , concede all $d \leq x$, and reject all $d > x$. Below x , the only types surviving equilibrium dominance are neutral types, if any; they induce concession at indifference. If none survives, the criterion imposes no restriction and the prior supports concession. Above x , a strictly rejecting posterior exists. If concession is sequentially rational under some posterior, monotonicity gives $U_C(d, c_1) \geq U_C(x, c_1)$ for every type, so all types survive and that rejecting belief is admissible. If concession is never sequentially rational, rejection is optimal under every posterior. Use a belief on neutral types when any survive, and

any belief otherwise. Resistance pays no type more than its equilibrium payoff. These beliefs also cover capped deviations and $x = 0$.

Finally, neutral types can use or mix over smaller accepted signals and larger rejected signals, provided the posterior strictly prefers rejection at each used rejected signal and follows Bayes' rule wherever applicable. At $x > 0$ these types always fight; at zero they can also operate after both responses. Their military actions and both payoffs therefore equal those under compliance with x . Only neutral types can randomize, and weighting the prior by signal probabilities leaves the acceptance numerator unchanged. If any neutral type exists, even a singleton, the criterion requires acceptance of every unused $d < x$. If none exists, it is silent there, although demands at or below the universal floor remain accepted. This establishes the claimed outcome equivalence without asserting uniqueness of signals or beliefs. $\square$

E.1. Zero Demands and Boundary Cases

A zero largest accepted demand exists exactly when $\bar{c} > p + a$ in the benchmark, and when $\bar{c} > s_R$ or $z > \lambda p$ in the operation game. The strict inequalities follow from the continuation tie rule. If $\bar{c} \leq p + a$, the benchmark interval is the singleton $\{p + c_2\}$: resistance always induces war, and compliance with $p + c_2$ gives the target $1 - p - c_2$ under every optimal continuation, so $\Phi(p + c_2) = 0$.

In the operation game, $x^e = 0$ exactly when $\ell \geq \bar{c}$. Every type then uses the same force action after either response to zero, so $\Phi(0) = 0$. If $\ell < \bar{c}$, a positive mass settles after zero but uses force after resistance, giving $\Phi(0) > 0$ and $x^e > 0$. When $x^e = 0$, every type may instead demand 1 and be rejected; compliance can have zero probability.

In the positive-floor region $\bar{c} \leq s_R$, $z \leq \lambda p$, the interval collapses exactly when $z = \lambda p$ and $p + c_2 = 2z$, yielding $\underline{x} = x^e = z$. If $z < \lambda p$, types in (p, b_R) fight after resistance but settle after compliance with $\underline{x}$, contributing $p + c_2 - \underline{x} > 0$. If $z = \lambda p$, then $b_R = \ell = p$: $p + c_2 < 2z$ gives strictly positive operation-to-settlement surplus above p , whereas $p + c_2 > 2z$ gives strictly positive war-to-operation surplus below p . Only $p + c_2 = 2z$ makes all contributions zero.

E.2. Minimum War and Conflict Cost

For $[v]_0^{\bar{c}} = \min\{\bar{c}, \max\{0, v\}\}$, the clipped war and total-force cutoffs in the operation game are

$$\kappa(x) = \left[\min\left\{p - x, \frac{p - x - z}{1 - \lambda}\right\} \right]_0^{\bar{c}} \quad \ell_x = [\max\{p - x, \ell\}]_0^{\bar{c}}$$

In the benchmark set $\kappa(x) = \ell_x = [p - x]_0^{\bar{c}}$. These formulas include zero demands, dominated operations, and support boundaries.

COROLLARY 2 (The Largest Demand Minimizes Conflict): Under Lemma 5's conditions, the equilibrium at x^e maximizes every coercer type's payoff and unconditional retained concession, and minimizes unconditional war probability and expected joint conflict cost. These conflict minima need not be unique.

Proof. Outcome equivalence gives unconditional war $W(x) = \frac{\kappa(x)}{\bar{c}}$ and retained concession $x(1 - W(x))$. Integrating military costs yields

$$K(x) = \frac{(1 - \lambda)\kappa(x)^2 + \lambda\ell_x^2}{2\bar{c}} + \frac{c_2\kappa(x)}{\bar{c}}$$

Both cutoffs weakly decrease with x , so W and K weakly decrease. Each coercer type's payoff and retained concession weakly increase. No type settles after actual resistance, since $a > 0$ makes available compliance strictly preferable. Expected joint surplus therefore equals $1 - K$ throughout the class. Evaluating at the largest feasible demand proves the result. $\square$

A selected increase in war or conflict cost therefore raises the minimum attainable level within the pure-response class. It does not rank every pair of equilibria across games. These are unconditional statements: moving neutral fighters to rejected signals preserves war and payoffs but changes war and retained concessions conditional on compliance. At zero, neutral operation users can also move; conditional quantities are undefined if compliance never occurs.

E.3. D1 and the Selected Pooling Equilibria

The stronger D1 test also compares types that survive equilibrium dominance. Let $q \in [0, 1]$ be the probability that the target concedes after x' . For each type, let $\mathcal{D}_{c_1}$ contain the values

of q that make the deviation strictly profitable and let $\mathcal{D}_{c_1}^0$ contain those that make it payoff-equivalent. D1 prunes c_1 if $\mathcal{D}_{c_1} \cup \mathcal{D}_{c_1}^0$ is a proper subset of $\mathcal{D}_{\hat{c}_1}$ for some other type $\hat{c}_1$.

In the benchmark, take $x' = x^\dagger + \varepsilon < p$ and define $k = p - x^\dagger$ and $k' = p - x'$. Types $c_1 \leq k'$ fight after either target response, so $\mathcal{D}_{c_1} = \emptyset$ and $\mathcal{D}_{c_1}^0 = [0, 1]$; D1 cannot prune them. Types in $(k', k]$ fight on the equilibrium path and after resistance but settle after concession of x' , so $\mathcal{D}_{c_1} = (0, 1]$ and $\mathcal{D}_{c_1}^0 = \{0\}$. Types above k require a strictly positive concession probability and are pruned by the switching types. The D1-admissible support is therefore $[0, k]$, not only the switching interval. The target is indifferent for the always-fighting types and strictly prefers concession for the switching types because $x' < p \leq p + c_2$. Under the maintained tie rule, every D1-admissible belief induces concession, and the switching types profit.

The same calculation applies in the operation game. Choose $x' = x^* + \varepsilon$ so that $x' + z < p$, and define $k = \kappa_W(x^*)$ and $k' = \kappa_W(x')$. Types below k' fight after either response and cannot be pruned. Types in $(k', k]$ fight on the equilibrium path and after resistance but use the operation after concession of x' . Types above k require a strictly positive concession probability and are pruned by the switching types. D1 therefore permits beliefs on $[0, k]$. Concession leaves the target indifferent for the always-fighting types and strictly improves its payoff for the switching types because $x' + z < p \leq p + c_2$. The tie rule again induces concession under every admissible belief, so the switching types profit.

If $c_2 > 0$, neither conclusion depends on the tie rule. In the benchmark choose $x' \in (p, p + c_2)$; in the operation game choose $x' \in (p - z, p + c_2 - z)$. Every type in the corresponding D1 support then switches away from war after concession, and the target strictly prefers concession. If $c_2 = 0$ and rejection is instead allowed at off-path indifference, a posterior on types that fight after either response can support rejection. Thus the pooling equilibria fail D1 under the maintained convention, but D1's conclusion at $c_2 = 0$ is tie-breaking-sensitive. The paper imposes the intuitive criterion rather than D1. [Proposition 9](#) extends this calculation to equilibria with pure target responses and a positive largest accepted demand.

A zero demand requires a separate comparison, because types can operate after either response without gaining from concession of zero.

PROPOSITION 9 (Positive Accepted Demands Fail D1 Under Pure Target Responses): Consider either game under [Assumption 1](#) and the rule that the target concedes when indifferent. Take any perfect Bayesian equilibrium in which the target's response is a pure function of the demand, and let $\bar{x} > 0$ be its largest accepted demand. Every coercer type's equilibrium payoff is $U_{C(\bar{x}, c_1)}$. If the war cutoff after concession of $\bar{x}$ is interior, the equilibrium fails D1. If $c_2 > 0$, this failure does not depend on the tie rule.

E.4. Proof of Proposition 9

Proof. For this proof, let $\kappa(x)$ denote the untruncated war cutoff after concession of x : it equals $\kappa_W(x)$ where the operation is active after that demand, $\kappa_S(x) = p - x$ where the operation is dominated, and $\kappa^\dagger(x) = p - x$ in the benchmark. On an uncapped local interval the branch expressions agree where the operation enters or leaves the envelope, so κ is continuous and strictly decreasing there. Lemma 2 gives $\kappa(x) \leq \kappa_R$, where κ_R is the resistance-stage war cutoff, so a type that fights after concession also fights after resistance. [Assumption 1\(ii\)](#) gives $\kappa(x) \leq \frac{p}{1-\lambda} < \bar{c}$, so at every demand a positive measure of types does not fight after concession.

The largest accepted demand and the equilibrium payoff. Conceding zero weakly improves the target's payoff under every posterior, as shown in the proof of Propositions 1 and 2, so the accepted set is nonempty. If it had no maximum, it would contain positive demands. The cap exclusion in Assumption 1's footnote and Lemma 2 then give a positive measure of nonfighting types whose payoffs strictly increase along accepted demands and exceed their payoff from any rejected demand. Those types would have no optimal demand. The accepted set therefore has a maximum. Under the proposition's hypothesis this maximum $\bar{x}$ is positive. Every type then has equilibrium payoff $u^*(c_1) = U_{C(\bar{x}, c_1)}$: a smaller accepted demand cannot pay more by monotonicity, a rejected demand cannot pay more by Lemma 2, and optimality rules out a lower payoff.

Pruning at an upward deviation. Let $k = \kappa(\bar{x})$ and suppose $0 < k < \bar{c}$. In the benchmark this implies $\bar{x} < p$. In the operation game it implies $\bar{x} + z < p$: on the active branch this follows directly from $k > 0$, while domination gives $z \leq \lambda(p - \bar{x}) < p - \bar{x}$. Choose $\varepsilon > 0$ small enough that $x' = \bar{x} + \varepsilon$ satisfies $k' = \kappa(x') > 0$, with $x' < p$ in the benchmark and $x' + z < p$ in the

operation game. These bounds keep the comparison uncapped, so κ is continuous and strictly decreasing from $\bar{x}$ to x' , giving $k' < k$. Let $q \in [0, 1]$ be the probability that the target concedes after x' . Type c_1 's payoff from the deviation is $qU_C(x', c_1) + (1 - q)U_R(c_1)$, so the deviation is strictly profitable exactly when $q\Delta(c_1) > E(c_1)$, where

$$\Delta(c_1) = U_C(x', c_1) - U_R(c_1), \quad E(c_1) = U_C(\bar{x}, c_1) - U_R(c_1)$$

Monotonicity gives $\Delta(c_1) \geq E(c_1) \geq 0$, and Lemma 2 gives $E(c_1) = 0$ exactly for $c_1 \leq k$ and $\Delta(c_1) = 0$ exactly for $c_1 \leq k'$. Three groups follow. Types $c_1 \leq k'$ fight after either response to x' , so $\mathcal{D}_{c_1} = \emptyset$ and $\mathcal{D}_{c_1}^0 = [0, 1]$. Types in $(k', k]$ have $\mathcal{D}_{c_1} = (0, 1]$ and $\mathcal{D}_{c_1}^0 = \{0\}$. Types above k have $E(c_1) > 0$, so $\mathcal{D}_{c_1} \cup \mathcal{D}_{c_1}^0 = [\frac{E(c_1)}{\Delta(c_1)}, 1]$ with $\frac{E(c_1)}{\Delta(c_1)} > 0$, a proper subset of $(0, 1]$. The first two groups have $\mathcal{D}_{c_1} \cup \mathcal{D}_{c_1}^0 = [0, 1]$, which no $\mathcal{D}_{\hat{c}_1}$ can properly contain, so D1 cannot prune them; the third group is pruned by the second, which has measure $\frac{k-k'}{c} > 0$. Every D1-admissible belief is therefore supported on $[0, k]$.

The target's reply. Under any belief on $[0, k]$, types in $[0, k']$ fight after concession of x' and after resistance, contributing $1 - p - c_2$ to both. Types in $(k', k]$ fight after resistance but not after concession of x' ; their target loss is $x' < p \leq p + c_2$ in the benchmark and at most $x' + z < p \leq p + c_2$ in the operation game. Each therefore raises the target's payoff from conceding above $1 - p - c_2$. Conceding is therefore weakly better than resisting, and strictly better when the belief places mass on $(k', k]$. Under the tie rule the target concedes.

Conclusion. If x' is off the equilibrium path, D1 confines the belief to $[0, k]$, the target concedes, and every type in $(k', k]$ receives $U_C(x', c_1) > U_R(c_1) = u^*(c_1)$, contradicting equilibrium. If x' is on the path, it is rejected because $\bar{x}$ is the largest accepted demand; every type that plays it must satisfy $U_R(c_1) = U_C(\bar{x}, c_1)$, hence $E(c_1) = 0$ and $c_1 \leq k$, so the belief obtained from Bayes' rule is also supported on $[0, k]$. The computation above then makes conceding weakly better, and the tie rule requires concession, contradicting rejection. Either way the equilibrium fails D1.

Independence of the tie rule. Suppose $c_2 > 0$ and take $x' \in (p - z, p + c_2 - z)$ in the operation game and $x' \in (p, p + c_2)$ in the benchmark. Since $\bar{x} < p$ in the benchmark and $\bar{x} + z < p$ in the operation game, each chosen x' exceeds $\bar{x}$. In the operation game $x' > p - z$ gives

$\lambda(p - x') < \lambda z < z$, so the operation is active after concession of x' and $\kappa(x') = \kappa_W(x') < 0$; in the benchmark $x' > p$ gives $\kappa^\dagger(x') < 0$. No admissible type then fights after concession of x' , every admissible type imposes loss $x' < p + c_2$ in the benchmark or at most $x' + z < p + c_2$ in the operation game, and the target strictly prefers conceding under every belief supported on $[0, k]$, whether that belief comes from D1 or from Bayes' rule. The switching types again profit.

A zero-demand exception in the operation game. The positive-demand condition is necessary. Take $p = 0.55$, $c_2 = 0.02$, $a = 0.30$, $\lambda = 0.08$, $z = 0.12$, and $\bar{c} = 2$, which satisfy the main closed-form premises. Let every type demand zero. After either response to zero the war cutoff is $\frac{p-z}{1-\lambda} \approx 0.467391$. After compliance, types up to $\frac{z}{\lambda} = 1.5$ then operate and higher types settle; after resistance, every nonfighting type operates. The target strictly prefers conceding zero, by $z\alpha = 0.03$.

Following any positive demand, assign a posterior concentrated on $c_1 = 1$. This type operates after both responses to zero and receives 0.04 at either history, so $E(1) = 0$. Any positive demand would strictly raise its payoff if conceded, giving $\mathcal{D}_1 = (0, 1]$ and $\mathcal{D}_1^0 = \{0\}$. No type profits from rejection relative to its equilibrium payoff, so no strict-gain set contains zero. D1 therefore cannot prune $c_1 = 1$. This type imposes target loss 0.12 after resistance. Its compliance loss is $x + 0.12$ for $x \leq 0.88$, 1 for $0.88 < x \leq 0.92$, and x above 0.92, each strictly greater than 0.12. Rejection of every positive demand is sequentially rational and deters deviations. The zero-demand pooling equilibrium consequently survives D1 despite its interior war cutoff. It also satisfies the intuitive criterion: the supporting type $c_1 = 1$ receives its equilibrium payoff from rejection and therefore cannot be eliminated by strict equilibrium dominance.

Such zero-demand equilibria do not exist for all parameters. Keep $p = 0.55$, $c_2 = 0.02$, $a = 0.30$, $\lambda = 0.08$, and $\bar{c} = 2$, but set $z = 0.04$. For every $0 < x < 0.04$, the operation is dominated after compliance because $x < p - \frac{z}{\lambda} = 0.05$. Types that fight after compliance also fight after resistance, leaving the target indifferent. Every other type settles after compliance but faces war or the operation after resistance, imposing loss 0.57 or 0.04, both greater than x . Concession is therefore weakly optimal under every posterior and required by the tie rule. No equilibrium can reject every positive demand.

A noninterior war cutoff. The interior-cutoff condition is also substantive. For example, if $p + a \geq \bar{c}$ in the benchmark, every type fights after resistance. A pooling equilibrium at $\bar{x} = p + c_2$ can survive D1: the target accepts all demands up to $p + c_2$, and the D1-admissible posterior following a larger demand is concentrated on $c_1 = 0$, against which the target strictly prefers resistance. Propositions 1 and 2 exclude this example through their interior war cutoffs. $\square$

F. EXTENSIONS AND SCOPE

F.1. Escalation Risk

Strict baseline reductions in coercive ability and increases in joint conflict cost persist for sufficiently small escalation risk when the baseline lies strictly within the active, interior region specified below. The every-type payoff-loss and war comparison also uses the stronger condition $x^\dagger > x^* + z$. Positive escalation risk makes settlement at the operation margin respond to the concession.

Suppose an operation triggers general war with probability ξ . General war reopens the dispute as elsewhere in the model. The coercer pays λc_1 whenever it uses the operation and pays c_1 in addition if escalation occurs. As before, $r = 1$ denotes resistance and $r = 0$ compliance. The operation lottery after transfer y gives expected payoffs

$$\begin{aligned} U_1^{L(y, c_1; \xi)} &= (1 - \xi)(y + S(y)) + \xi p - (\lambda + \xi)c_1, \\ U_2^{L(y; \xi)} &= (1 - \xi)(1 - y - S(y)) + \xi(1 - p - c_2). \end{aligned} \tag{21}$$

Settlement pays the coercer $y - a\mathbb{1}[r = 1]$, and war pays $p - c_1$. For $\xi \in [0, 1 - \lambda)$, the slopes of war, the operation, and settlement remain ordered. On an active three-action branch, the coercer fights below $\kappa_W(y; \xi)$, uses the operation between the two cutoffs, and settles above $\kappa_L(y, r; \xi)$, where

$$\begin{aligned} \kappa_W(y; \xi) &= (1 - \xi) \frac{p - y - S(y)}{1 - \lambda - \xi}, \\ \kappa_L(y, r; \xi) &= \frac{(1 - \xi)S(y) + \xi(p - y) + a\mathbb{1}[r = 1]}{\lambda + \xi}. \end{aligned} \tag{22}$$

At an accepted demand, [Assumption 1\(i\)](#) gives $S(y) = z$ and

$$\frac{\partial \kappa_L(y, 0; \xi)}{\partial y} = -\frac{\xi}{\lambda + \xi}. \quad (23)$$

Thus any positive escalation risk makes a larger concession increase settlement on the operation–settlement margin. Setting $\xi = 0$ recovers (6) and the demand-invariant cutoff in Proposition 3.

Define $W_\xi(y, r)$ as the probability of choosing war directly and $L_\xi(y, r)$ as the probability of choosing either war or the operation under the coercer's globally optimal continuation choice. On an active branch, $W_\xi(y, r) = F(\kappa_W(y; \xi))$ and $L_\xi(y, r) = F(\kappa_L(y, r; \xi))$. If the operation is dominated, $W_\xi(y, r) = L_\xi(y, r) = F(p - y + a\mathbb{1}[r = 1])$. Write $W_{\xi,R} = W_\xi(0, 1)$ and $L_{\xi,R} = L_\xi(0, 1)$ after resistance, and $W_\xi^{C(x)} = W_\xi(x, 0)$ and $L_\xi^{C(x)} = L_\xi(x, 0)$ after compliance. The corresponding probabilities of general war are

$$\begin{aligned} R_{\xi,R} &= W_{\xi,R} + \xi(L_{\xi,R} - W_{\xi,R}), \\ R_\xi^{C(x)} &= W_\xi^{C(x)} + \xi[L_\xi^{C(x)} - W_\xi^{C(x)}]. \end{aligned} \quad (24)$$

The target's expected losses after resistance and compliance are

$$\begin{aligned} D_{\xi,R} &= (p + c_2)R_{\xi,R} + (1 - \xi)z(L_{\xi,R} - W_{\xi,R}), \\ D_\xi^{C(x)} &= x[1 - R_\xi^{C(x)}] + (p + c_2)R_\xi^{C(x)} + (1 - \xi)S(x)[L_\xi^{C(x)} - W_\xi^{C(x)}]. \end{aligned} \quad (25)$$

The target concedes exactly when $D_\xi^{C(x)} \leq D_{\xi,R}$. Hence the selected equilibrium demand and coercive ability are

$$x_\xi = \max\left\{x \in [0, 1] : D_\xi^{C(x)} \leq D_{\xi,R}\right\}, \quad \Omega_\xi = x_{\xi \left[1 - R_\xi^{C(x_\xi)}\right]}. \quad (26)$$

The active, interior region also admits a closed form. Suppose $S(y) = z$, the two post-compliance cutoffs and the post-resistance war cutoff are interior, the operation is active after both responses, and $L_{\xi,R} = 1$. Define

$$\begin{aligned}
A_\xi &= 1 - \lambda - \xi, \quad d_\xi = \lambda + \xi, \quad t_\xi = 1 - \xi, \\
w_\xi &= \frac{t_\xi(p-z)}{A_\xi \bar{c}}, \quad \ell_\xi = \frac{t_\xi z + \xi p}{d_\xi \bar{c}}, \\
r_\xi &= \xi \ell_\xi + t_\xi w_\xi, \beta_\xi = \frac{\xi^2}{d_\xi \bar{c}} + \frac{t_\xi^2}{A_\xi \bar{c}}, \\
\eta_\xi &= \frac{t_\xi}{A_\xi \bar{c}} - \frac{\xi}{d_\xi \bar{c}}.
\end{aligned} \tag{27}$$

Then $W_\xi^{C(x)} = w_\xi - t_\xi \frac{x}{A_\xi \bar{c}}$, $L_\xi^{C(x)} = \ell_\xi - \xi \frac{x}{d_\xi \bar{c}}$, and $R_\xi^{C(x)} = r_\xi - \beta_\xi x$. Let

$$\begin{aligned}
H_\xi &= \xi(p + c_2) + t_\xi(p + c_2)w_\xi + t_\xi z(1 - w_\xi), \\
C_\xi &= H_\xi - [(p + c_2)r_\xi + t_\xi z(\ell_\xi - w_\xi)], \\
B_\xi &= 1 - r_\xi - (p + c_2)\beta_\xi + t_\xi z\eta_\xi.
\end{aligned} \tag{28}$$

The binding acceptance condition is

$$\beta_\xi x^2 + B_\xi x - C_\xi = 0. \tag{29}$$

When $C_\xi > 0$ and the target's acceptance margin is strictly decreasing, the selected demand is the unique positive root

$$x_\xi = \frac{-B_\xi + \sqrt{B_\xi^2 + 4\beta_\xi C_\xi}}{2\beta_\xi}. \tag{30}$$

At $\xi = 0$, multiplying (29) by $(1 - \lambda)\bar{c}$ yields the quadratic for (9).

PROPOSITION 10 (Local Robustness to Escalation Risk): Suppose the operation-game closed-form conditions and the inequalities defining its active, interior regime hold strictly at $\xi = 0$. Then there is an $\bar{\xi} > 0$ such that the selected pooling equilibrium in (30) exists and varies continuously for every $\xi \in [0, \bar{\xi})$. Every strict comparison below survives on this interval: (i) if Proposition 4 gives $\Omega^* < \Omega^\dagger$, then $\Omega_\xi < \Omega^\dagger$; (ii) if the stronger sufficient condition $x^\dagger > x^* + z$ holds, every coercer type remains weakly worse off with the operation, a positive measure remains strictly worse off, and both direct and total general war after compliance remain strictly more likely than in the benchmark; (iii) if Proposition 5 or Proposition 7 gives $K^* > K^\dagger$, then escalation

preserves the higher total conflict cost; and (iv) if Proposition 7 gives $\frac{\partial K^*}{\partial \lambda} < 0$, then $\frac{\partial K_\xi^*}{\partial \lambda} < 0$.

Proof. The proof combines a local argument with compactness. Let $G_{\xi(x)} = D_{\xi,R} - D_\xi^{C(x)}$. The strict active, interior conditions give a $\delta > 0$ such that $[x^* - \delta, x^* + \delta]$ remains on the same uncapped branch for all sufficiently small ξ . On this interval, G_ξ is continuously differentiable in (x, ξ) , and [Assumption 1\(ii\)](#) gives $G_0'(x^*) < 0$. After shrinking δ and the neighborhood of zero, $\frac{\partial G_{\xi(x)}}{\partial x} < 0$ throughout the interval. The implicit function theorem therefore supplies a unique local root x_ξ , continuous in ξ . Proposition 2 establishes that x^* is the unique largest accepted demand at $\xi = 0$, so $G_0(x) < 0$ on the compact set $[x^* + \delta, 1]$. Joint continuity of the optimal-continuation probabilities implies that G_ξ converges uniformly to G_0 on this set. Hence $G_{\xi(x)} < 0$ there for all sufficiently small ξ . The negative local derivative rules out another accepted demand between x_ξ and $x^* + \delta$, while compactness rules out one above that interval. Thus x_ξ remains the globally largest accepted demand without requiring global monotonicity away from the selected branch.

Concession also continues to support pooling. For each type, war pays the same after either response, settlement rises from $-a$ to x , and the operation payoff rises by $(1 - \xi)[x + S(x) - z] \geq 0$. The coercer's optimized continuation payoff is therefore weakly higher after concession. Settlement types strictly prefer a larger accepted demand. Operation users weakly prefer it globally and strictly prefer it on the local uncapped branch, where $S(x) = z$. Fighting types are indifferent. The optimized payoff $U_{C,\xi}(x, c_1) = \max\{p - c_1, x, (1 - \xi)[x + S(x)] + \xi p - (\lambda + \xi)c_1\}$ is weakly increasing over the entire demand set. The preceding compactness argument identifies x_ξ as the largest prior-accepted demand for sufficiently small ξ . These properties satisfy [SI §C.1. Off-Path Beliefs and the Intuitive Criterion](#), so the selected pooling equilibrium satisfies the criterion throughout that neighborhood.

Strict regime inequalities remain satisfied in a neighborhood of $\xi = 0$. On that neighborhood, [\(29\)](#) has $\beta_\xi > 0$, and its positive root is continuous in ξ . Thus $x_\xi \rightarrow x^*$, the cutoffs in [\(22\)](#) are continuous, and $W_\xi^{C(x_\xi)}$, $L_\xi^{C(x_\xi)}$, $R_\xi^{C(x_\xi)}$, and Ω_ξ are continuous at zero. Any strict coercive-ability comparison at zero therefore persists for sufficiently small ξ , proving (i).

For (ii), continuity of the demand gives $x^\dagger > x_\xi + z$ for all sufficiently small ξ . The coercer's post-compliance payoff in the operation game is

$$U_{\xi(c_1)} = \max\{p - c_1, x_\xi, (1 - \xi)(x_\xi + z) + \xi p - (\lambda + \xi)c_1\}.$$

The benchmark payoff is $U^{\dagger(c_1)} = \max\{p - c_1, x^\dagger\}$. Settlement is lower in the operation game. The operation payoff can be written as $(1 - \xi)(x_\xi + z) + \xi(p - c_1) - \lambda c_1$, which is no greater than $U^{\dagger(c_1)}$ because $x_\xi + z < x^\dagger$. War pays $p - c_1$ in both games. Hence $U_{\xi(c_1)} \leq U^{\dagger(c_1)}$ for every type, with strict inequality for the positive measure of sufficiently high-cost types that settle in both games. Proposition 5 gives $W_0^{C(x^*)} > F(\kappa^\dagger(x^\dagger))$. Continuity preserves this strict direct-war comparison, and (24) implies the same strict comparison for total general war.

For (iii), expected conflict cost on the active branch is

$$K_\xi = \int_0^{\kappa_W(x_\xi; \xi)} (c_1 + c_2) dF(c_1) + \int_{\kappa_W(x_\xi; \xi)}^{\kappa_L(x_\xi, 0; \xi)} [\lambda c_1 + \xi(c_1 + c_2)] dF(c_1).$$

Its limits and integrands are continuous in ξ , so $K_\xi \rightarrow K^*$. Every strict $K^* > K^\dagger$ comparison in Proposition 5 or Proposition 7 therefore persists for sufficiently small escalation risk.

Finally, the binding equation in (29) has a nonzero derivative with respect to x at the selected root. The implicit function theorem makes $x_{\xi(\lambda)}$ continuously differentiable in (ξ, λ) within the strict regime. The cutoffs, integrands, and integral limits defining K_ξ are continuously differentiable there, so $\frac{\partial K_\xi^*}{\partial \lambda}$ is continuous at $\xi = 0$. A strict negative derivative in Proposition 7 remains negative for sufficiently small ξ , proving (iv). Taking the minimum of the neighborhoods established for the regime and the four comparisons gives $\bar{\xi} > 0$ and proves the proposition. $\square$

The proposition establishes robustness to sufficiently small escalation risk. Its every-type payoff comparison uses the stronger condition $x^\dagger > x^* + z$; the weak payoff comparison at equality in (5) alone does not establish robustness. It does not order outcomes when escalation is large enough to change the continuation regime.

F.2. Power Shift and Partial Recontestation

Under the active, interior conditions stated below, a positive power shift preserves the operation–settlement comparison while reducing how much a concession lowers the war cutoff. If

the accepted demand falls, the every-type payoff-loss and war comparisons then have different thresholds.

The baseline makes the probability of victory independent of an earlier transfer. Some concessions instead change the subsequent balance of power. Suppose compliance converts a fraction γ of the concession into the coercer's probability of victory. Nothing has been transferred after resistance, so that node is unchanged and the extension operates on the post-compliance branch alone. The following closed forms apply at values of γ for which the benchmark war cutoffs remain interior, the operation remains active with interior post-compliance cutoffs, $L_R = 1$, and $S(x) = z$ at the selected demands. These conditions must be checked in the shifted game; their validity at $\gamma = 0$ alone is insufficient. Outside this branch, the demand is determined by the acceptance condition using the globally optimal continuation menu.

REMARK 5 (Power Shift): Let $\gamma \in [0, 1 - p]$ denote the share of each conceded unit converted into military advantage. War after compliance then pays the coercer $p + \gamma x - c_1$ and the target $1 - p - c_2 - \gamma x$. If war reopens a share ρ of the transfer and contests the rest of the good, then $\gamma = (1 - \rho)(1 - p)$. The baseline is $\gamma = 0$; at $\gamma = 1 - p$, every post-compliance action leaves the transfer in place.

After compliance, the war cutoffs are

$$\kappa^\dagger(x) = p - (1 - \gamma)x, \quad \kappa_W(x) = \frac{p - (1 - \gamma)x - z}{1 - \lambda}$$

while $\kappa_L(x) = \frac{z}{\lambda}$ and $\alpha = 1 - F(\frac{z}{\lambda})$ are unchanged, as is the entire post-resistance node.

Define the target's effective payment as $P = x[1 - (1 - \gamma)W(x)]$. It still equals $\Theta^* - \Delta^*$ for Θ^* and Δ^* as defined in (3). The largest accepted demands are the positive roots of (31), and their effective payments are

$$P^\dagger = \frac{(p + c_2)((1 - \gamma)x^\dagger + a)}{\bar{c}}, \quad P^* = \frac{(1 - \gamma)x^*(p + c_2 - z)}{(1 - \lambda)\bar{c}} + z\alpha.$$

Definition 1 instead gives coercive ability as

$$\Omega^\dagger = x^\dagger \left[1 - \frac{p - (1 - \gamma)x^\dagger}{\bar{c}} \right],$$

$$\Omega^* = x^* \left[1 - \frac{p - (1 - \gamma)x^* - z}{(1 - \lambda)\bar{c}} \right].$$

The capability lowers coercive ability exactly when $\Omega^\dagger > \Omega^*$ under these formulas.

Setting $\gamma = 0$ recovers (8), (9), and Proposition 4.

$$(1 - \gamma)^2 x^2 + (\bar{c} - (1 - \gamma)(2p + c_2))x - (p + c_2)a = 0,$$

$$(1 - \gamma)^2 x^2 + ((1 - \lambda)\bar{c} - (1 - \gamma)(2p + c_2) + 2(1 - \gamma)z)x - (1 - \lambda)\bar{c}z\alpha = 0 \tag{31}$$

The cutoffs follow from the payoffs. After compliance, war pays $p + \gamma x - c_1$, settlement pays x , and the operation pays $x + z - \lambda c_1$, so war beats settlement when $c_1 < p - (1 - \gamma)x$ and beats the operation when $(1 - \lambda)c_1 < p - (1 - \gamma)x - z$. Neither settlement nor the operation

involves war, so their comparison is the baseline's and Proposition 3 applies unchanged: the transfer cancels, κ_L is constant in the demand, and compliance still buys no restraint on the operation margin. What the transfer buys is war avoidance, and $1 - \gamma$ is the rate at which it does so. The target's loss from complying is $x(1 - W(x)) + (p + c_2 + \gamma x)W(x) + z(L(x) - W(x))$, which equals $x(1 - (1 - \gamma)W(x)) + (p + c_2)W(x) + z(L(x) - W(x))$. Equating this to the unchanged loss from resisting and substituting $W_R - W(x) = (1 - \gamma)\frac{x}{(1 - \lambda)\bar{e}}$ gives the second quadratic; removing the operation and using $W = L = F(\kappa^\dagger)$ gives the first. The [Assumption 1\(ii\)](#) is unchanged and remains sufficient, because it binds most tightly at $\gamma = 0$.

These demands also support pooling under the intuitive criterion. The optimized compliance payoff is $U_{C,\gamma}(x, c_1) = \max\{p + \gamma x - c_1, x, \min\{x + z, 1\} - \lambda c_1\}$, with the operation term omitted in the benchmark. It is weakly increasing throughout the demand set, and each continuation action pays weakly more after concession than after resistance. Apply [SI §C.1. Off-Path Beliefs and the Intuitive Criterion](#) to the largest prior-accepted demand in each game. When $\gamma > 0$, every type strictly loses from a smaller demand relative to the positive pooling demand, whatever the target's response: all compliance payoffs increase strictly on the uncapped interval up to that demand, and resistance also pays strictly less. Every type is then equilibrium dominated at that deviation, so the criterion imposes no belief restriction there.

Two comparisons change. The demand enters the target's effective-payment term through $(1 - \gamma)x$, while coercive ability remains $x(1 - W)$. War after compliance also responds less to the demand, because the post-compliance war cutoff now moves by $1 - \gamma$ rather than by one for each unit conceded.

REMARK 6 (Payoffs and War under a Power Shift): Fix $\gamma \in [0, 1 - p]$ and maintain the active, interior conditions of the preceding remark, including an active operation after compliance. Every coercer type is weakly worse off with the operation if and only if $x^\dagger \geq x^*$ and (32) holds, and war after compliance is strictly more likely if and only if (33) holds. When $\gamma > 0$ and $x^\dagger > x^*$, the second condition is the stronger of the two. When $\gamma > 0$ and $x^\dagger > x^*$, the coercer's war payoff after compliance rises in the transfer, so types that fight in both games lose $\gamma(x^\dagger - x^*)$. If $\gamma > 0$, $x^\dagger > x^*$, and (32) holds strictly, the payoff loss is strict for almost every type. At $\gamma = 0$ types that fight in both games are indifferent, and the two exact conditions have the same left-hand side, with a weak inequality for the payoff comparison and a strict inequality for the war comparison. They recover Proposition 5's weak frontier and strict-war comparison on the maintained active branch.

$$x^\dagger + \lambda \kappa^\dagger(x^\dagger) \geq x^* + z \quad (32)$$

$$(1 - \gamma)(x^\dagger - x^*) + \lambda \kappa^\dagger(x^\dagger) > z \quad (33)$$

Both conditions compare the operation-game payoff with $U^{\dagger(c_1)} = \max\{p + \gamma x^\dagger - c_1, x^\dagger\}$. An interior settlement region in both games makes $x^\dagger \geq x^*$ necessary for every-type weak loss. Given that inequality, benchmark war weakly dominates operation-game war, and benchmark settlement weakly dominates operation-game settlement. It remains to compare the operation payoff $x^* + z - \lambda c_1$. The benchmark payoff advantage over the operation decreases up to $\kappa^\dagger(x^\dagger)$ and increases afterward. The smallest benchmark advantage therefore occurs at that cutoff and is $x^\dagger + \lambda \kappa^\dagger(x^\dagger) - x^* - z$. Nonnegativity gives (32) and proves the payoff equivalence. Rearranging $\kappa_W(x^*) > \kappa^\dagger(x^\dagger)$ gives (33). At $\gamma = 0$, write $d = x^\dagger - x^*$ and $b = p - x^\dagger$. Every-type weak loss requires $d \geq 0$ and $d + \lambda b \geq z$, whereas strict war requires $d + \lambda b > z$ on this active branch. These are the corresponding cases of Proposition 5. The stronger condition $d > z$ in (18) is sufficient for both and is unnecessary.

F.3. The Operation's Audience Cost

An audience cost on the operation makes it less attractive after either target response. The conditions below determine whether it remains in the continuation menu. Within the stated active, closed-form branch, a higher audience cost on the operation raises the accepted demand and coercive ability when $z < p + c_2$.

REMARK 7 (An Audience Cost on the Operation): Suppose using the operation imposes an audience cost $a_L \in [0, a]$ after either response, in addition to λc_1 . After resistance, $\kappa_{L,R} = \frac{z+a-a_L}{\lambda}$ and $\kappa_{W,R} = \frac{p-z+a_L}{1-\lambda}$. A positive measure of types operates exactly when $\max\{0, \kappa_{W,R}\} < \min\{\bar{c}, \kappa_{L,R}\}$. If the benchmark resistance cutoff is interior, this condition, and strict war reduction, are equivalent to $z + a - a_L > \lambda(p + a)$.

After compliance with $S(x) = z$, the cutoffs are $\kappa_L(x) = \frac{z-a_L}{\lambda}$ and $\kappa_W(x) = \frac{p-x-z+a_L}{1-\lambda}$. Under Assumption 1(ii), positive-measure operation use requires and is implied by $a_L < z$ and $a_L < z - \lambda(p - x)$. On this active branch $\alpha = 1 - F(\frac{z-a_L}{\lambda})$; otherwise the operation is dominated on the supported types and $\alpha = 1 - F(p - x)$.

On the active closed-form branch, maintain interior war cutoffs and an interior post-compliance operation-settlement cutoff, full reach, and $L_R = 1$. The last condition requires $a_L \leq z + a - \lambda\bar{c}$. The binding acceptance condition is (34), and x^* and Ω^* strictly increase in a_L within this branch when $z < p + c_2$, with the corresponding one-sided comparison at a boundary. Setting $a_L = 0$ recovers the baseline on this branch.

$$x^2 + x((1 - \lambda)\bar{c} - 2p - c_2 + 2z - a_L) - (1 - \lambda)\bar{c}z\alpha = 0 \quad (34)$$

Settlement after resistance pays $-a$, the operation pays $z - \lambda c_1 - a_L$, and war pays $p - c_1$. Their intersections give the two resistance cutoffs. The inequality $\kappa_{W,R} < \kappa_{L,R}$ reduces to $\lambda(p + a) < z + a - a_L$, as does $\kappa_{W,R} < p + a$. With $0 < p + a < \bar{c}$, the interval where the operation replaces war has positive supported mass exactly when this inequality holds. Without that support condition, ordered cutoffs above $\bar{c}$ need not change any type's action.

After compliance, settlement pays x , the operation $x + z - \lambda c_1 - a_L$, and war $p - c_1$. If $a_L \geq z$, settlement weakly dominates the operation for all supported types. If $a_L < z$, the operation–settlement cutoff is positive and the war–operation cutoff is below $\frac{p}{1-\lambda} < \bar{c}$. The remaining activity condition is therefore the ordering of the two cutoffs, equivalent to $a_L < z - \lambda(p - x)$. These comparisons establish the two branches of the settlement probability.

In the active closed-form region, $L(x) = \frac{z-a_L}{\lambda \bar{c}}$ and $L_R = 1$, while $W_R - W(x) = \frac{x}{(1-\lambda)\bar{c}}$ because a_L shifts both war cutoffs equally. The substitution used for [Proposition 4](#) gives $\Omega^* = \frac{x^*(p+c_2-z)}{(1-\lambda)\bar{c}} + z\alpha$ at the positive root of [\(34\)](#). Implicit differentiation gives

$$\frac{\partial x^*}{\partial a_L} = \frac{x^* + (1-\lambda)\frac{z}{\lambda}}{2x^* + (1-\lambda)\bar{c} - 2p - c_2 + 2z - a_L} > 0$$

so Ω^* rises with a_L on this branch when $z < p + c_2$. Once the operation is dominated after compliance, neither [\(34\)](#) nor its formula for α applies; the comparison follows the war–settlement branch and no global ordering follows from the active-branch derivative.

To verify that this root is the largest demand accepted under the prior, differentiate the target’s compliance loss. On an active, uncapped branch with a supported war cutoff, its derivative is $1 - \frac{2p+c_2-2x-2z+a_L}{(1-\lambda)\bar{c}} > 0$. On a dominated branch it is $1 - \frac{2p+c_2-2x}{\bar{c}} > 0$, and after war disappears it is 1. These signs follow from $a_L < z$ and Assumption 1(ii), and the loss is continuous at branch boundaries. Capped demands are strictly rejected: $1 - a_L > p$ makes the operation dominate war there, so compliance costs at least $x > 1 - z \geq \max\{p + c_2, z\}$, the upper bound on resistance loss. The binding root is therefore the largest prior-accepted demand.

The selected demand on this branch also satisfies the intuitive criterion. Its optimized compliance payoff is $U_{C,a_L}(x, c_1) = \max\{p - c_1, x, \min\{x + z, 1\} - \lambda c_1 - a_L\}$, which is weakly increasing in the demand. Because the operation incurs the same audience cost after either response, concession weakly improves every continuation payoff relative to resistance. Thus [§C.1. Off-Path Beliefs and the Intuitive Criterion](#) applies to the largest demand accepted under the prior, with the closed form restricted to the stated active branch.